Anya DeCarlo
Biostats 208 Final

QUESTION #1

The dataset `final-hersdata.dta` is a subset of data from the HERS trial.

Focus on the outcome variable `HDL` (high-density lipoprotein in *mg/dL*), and the following three predictors: physical activity `physact`, `age` and body mass index `BMI`. The variable `physact` is coded 1: much less active, 2: somewhat less active (labeled as "less" in the dataset), 3: about as active (labeled as "equally" in the dataset), 4: somewhat more active, 5: much more active.

Question 1.1:
(i) How would you classify the data type of the predictor `physact`?

I would classify the data type `physact` as an ordinal data type.

(*ii*) Is a test for linear trend appropriate for this predictor? Why?

Yes, a test for a linear trend is appropriate for this predictor. The levels of `physact` are equally spaced and have a numeric order to them.

(*iii*) Fit a regression model with `HDL` as the outcome and `physact` as a predictor.

```
hersLM <- lm(HDL ~ physact, data = hers)
```

```
MODEL INFO:
Observations: 276
Dependent Variable: HDL
Type: OLS linear regression

MODEL FIT:
F(4,271) = 5.77, p = 0.00
R² = 0.08
Adj. R² = 0.06

Standard errors:OLS
```

	Est.	S.E.	t val.	p
(Intercept)	44.94	3.53	12.74	0.00
physactsomewhat less active	-0.07	4.03	-0.02	0.99
physactabout as active	7.11	3.90	1.82	0.07
physactsomewhat more active	10.18	3.84	2.65	0.01
physactmuch more active	10.26	4.30	2.39	0.02

(*iv*) Perform a test for linear trend. (Evaluating departure from linear trend is optional.)

```
hers$physact.cont <- hers$physact
contrasts(hers$physact.cont) <- contr.poly(nlevels(hers$physact))
hersLM <- lm(HDL ~ physact.cont, data = hers)
jtools::summ(hersLM, confint = TRUE, digits = 3)
```

Observations: 276
Dependent Variable: HDL
Type: OLS linear regression

MODEL FIT:
$F(4,271) = 5.767$, $p = 0.000$
$R^2 = 0.078$
Adj. $R^2 = 0.065$

Standard errors:OLS

	Est.	2.5%	97.5%	t val.	p
(Intercept)	50.438	48.378	52.497	48.215	0.000
physact.cont.L	9.728	4.156	15.301	3.437	0.001
physact.cont.Q	-1.020	-6.043	4.002	-0.400	0.690
physact.cont.C	-3.236	-7.315	0.843	-1.562	0.120
physact.cont^4	1.490	-1.962	4.942	0.850	0.396

(*v*) Write 2-4 sentences describing, as you would in a paper, the estimated effect of physical activity on HDL levels. (Include 95% CIs when appropriate!)

A linear regression model was fit with HDL as the outcome and physical activity (`physact`) as the sole predictor. Although `physcact` is stored as a categorical variable, its five levels are ordinal and therefore orthogonal polynomial contrasts can be appl 2.1 to investigate linear, quadratic, cubic, and higher – order trends. These results show a statistically significant linear trend between physical activity and HDL ((β = 9.728, 95% CI: 4.156 to 15.301, p = 0.001). The estimated effect of physical activity can therefore be described as a positive linear association, where a one level increase in reported physical activity level is associated with a 9.7 mg/dl increase in HDL.

Question 1.2: **(*i*)** Assume that it is reasonable to consider age and BMI as possible confounding influences on the relationship between physical activity and HDL. Adjust the relationship in Q1.1 for the variables `age` and `BMI` by including them as predictors in the regression model. (You may assume that these variables can be included as linear terms.) Interpret the results, and answer the following question:

```
hersLM_Adjusted <- lm(HDL ~ physact + age + BMI, data = hers)
```

MODEL INFO:
Observations: 276
Dependent Variable: HDL
Type: OLS linear regression

MODEL FIT:
$F(6,269) = 10.45$, $p = 0.00$
$R^2 = 0.19$
Adj. $R^2 = 0.17$

Standard errors:OLS

	Est.	S.E.	t val.	p
(Intercept)	42.74	11.50	3.72	0.00
physactsomewhat less active	1.76	3.81	0.46	0.64
physactabout as active	6.93	3.69	1.88	0.06
physactsomewhat more active	8.40	3.63	2.31	0.02
physactmuch more active	7.42	4.08	1.82	0.07
age	0.36	0.13	2.74	0.01
BMI	-0.75	0.17	-4.38	0.00

(*ii*) Are the results consistent with `age` and `BMI` having a confounding influence on the relationship between `physact` and `HDL`?
Yes, the results are consistent with age and BMI acting as confounders in the relationship between physical activity and HDL. In the unadjusted model, women who reported being much more active had HDL levels

10.26 mg/dL higher than those who were reported being “much less active” (p = 0.02). In the adjusted model, women who reported being much more active had HDL levels 7.42 mg/dL higher than those were reported being “much less active” (p = 0.07). The effect not only decreased in the adjusted model but is not statistically significant at the alpha = 0.05 level. Therefore, the unadjusted estimate shows a more significant effect than the adjusted estimate, and after analysis we find evidence that age and BMI confound the unadjusted association.

(*iii*) Write 2-4 sentences describing the adjusted association between `physact` and `HDL`.

The adjusted estimate shows that average HDL increases 10.26 mg/DL among women who report being much more active compared to those who report being much less active (95% CI: 1.63 – 18.88 mg/dL, p = 0.02). We also find estimated effect of physical activity can still be described as a positive linear association, where a one level increase in reported physical activity level is associated with a 6.791 mg/dl increase in HDL (p = 0.013).

```
# Question 1.2 (i): Adjust for age and BMI
# Fit adjusted model
hersLM_Adjusted <- lm(HDL ~ physact + age + BMI, data = hers)
jtools::summ(hersLM_Adjusted, confint = TRUE, digits = 3)

# Compare unadjusted and adjusted models to assess confounding
# Extract coefficients for comparison
unadj_coefs <- coef(hersLM)[grep("physact", names(coef(hersLM)))]
adj_coefs <- coef(hersLM_Adjusted)[grep("physact", names(coef(hersLM_Adjusted)))]

# Calculate percent change in coefficients
percent_change <- (adj_coefs - unadj_coefs) / abs(unadj_coefs) * 100
print(data.frame(
  Unadjusted = unadj_coefs,
  Adjusted = adj_coefs,
  Percent_Change = percent_change

))

# Encode physact as orthogonal polynomial contrasts
hers$physact.cont <- hers$physact
contrasts(hers$physact.cont) <- contr.poly(nlevels(hers$physact))

# Fit the adjusted model with physact as a continuous term
hersLM_adjusted_trend <- lm(HDL ~ physact.cont + age + BMI, data = hers)

# Summarize the model
jtools::summ(hersLM_adjusted_trend, confint = TRUE, digits = 3)
```

MODEL INFO:
Observations: 276
Dependent Variable: HDL
Type: OLS linear regression

MODEL FIT:
$F(6,269) = 10.448$, $p = 0.000$
$R^2 = 0.189$
Adj. $R^2 = 0.171$

Standard errors:OLS

	Est.	2.5%	97.5%	t val.	p
(Intercept)	47.641	25.659	69.622	4.267	0.000
physact.cont.L	6.791	1.458	12.124	2.507	0.013
physact.cont.Q	−2.453	−7.218	2.311	−1.014	0.312
physact.cont.C	−1.850	−5.722	2.021	−0.941	0.348
physact.cont^4	1.001	−2.271	4.274	0.602	0.547
age	0.359	0.101	0.618	2.737	0.007
BMI	−0.753	−1.092	−0.415	−4.380	0.000

Question 1.3: **(*i*)** Based on the model in Q1.2, estimate the mean difference in `HDL` between a person who is ”much more” active compared with someone who is ”about as” active (adjusting for `age` and `BMI`).

The estimated difference in average HDL between a person who is “much” more active compared with someone who is “about as” active adjusting for age and BMI is 0.49 mg/dL.

```
hypotheses(hersLM_Adjusted, "`physactmuch more active` - `physactabout as
active`= 0", conf_level = 0.95)
```

```
 Hypothesis Estimate Std. Error     z
  `physactmuchmoreactive`-`physactabouta
 sactive`=0    0.489       2.84 0.172
  Pr(>|z|)   S 2.5 % 97.5 %
     0.863 0.2 -5.07   6.05
```

(*ii*) Give the 95% confidence interval for this effect

The estimated difference in average HDL between a person who is “much” more active compared with someone who is “about as” active adjusting for age and BMI is 0.49 mg/dL (95% CI: -5.07 – 6.05, p = 0.863) and are not statistically significant at the alpha = 0.05 level.

Question 1.4: **(*i*)** Based on the model in Q1.2, estimate the effect on mean `HDL` associated with a 10-year *decrease* in age (adjusting for `BMI` and `physact`).

The estimated effect on mean HDL associated with a 10-year decrease in age after adjusting for BMI and physact is – 3.59 mg/dL.

```
hypotheses(hersLM_Adjusted, "`age` * -10 = 0", conf_level = 0.95)
```

```
> hypotheses(hersLM_Adjusted, "`age` *
-10 = 0", conf_level = 0.95)

  Hypothesis Estimate Std. Error     z
Pr(>|z|)   S 2.5 % 97.5 %
 `age`*-10=0    -3.59       1.31 -2.74
  0.0062 7.3 -6.16  -1.02
```

(*ii*) Give the 95% confidence interval for this effect.

The estimated effect on mean HDL associated with a 10- year decrease in age after adjusting for BMI and phyact is -3.59 mg/dL. We find that a 10 year decrease in age is associated with a 3.59 mg/dL increase in HDL (95% CI 1.02 – 6.16, p = 0.0062). These results are statistically significant at the alpha = 0.05 level.

Question 1.5: **(*i*)** Based on the model in Q1.2, estimate the effect on mean `HDL` of a 5-unit increase in `BMI` (adjusting for `age` and `physact`).

```
hypotheses(hersLM_Adjusted, "`BMI` * 5 = 0", conf_level = 0.95)
```

```
> #  5-unit increase in BMI (adjusting f
or age and physact).
> hypotheses(hersLM_Adjusted, "`BMI` * 5
 = 0", conf_level = 0.95)

 Hypothesis Estimate Std. Error     z Pr
(>|z|)    S 2.5 % 97.5 %
  `BMI`*5=0    -3.77       0.86 -4.38
<0.001 16.4 -5.45  -2.08
```

(*ii*) Give the 95% confidence interval for this effect.

The estimated effect of a 5 unit increases in BMI (adjusting for age and physact) is -3.77 mg/dL indicating that a 5- unit increase in BMI is associated with a 3.77 mg/dL decrease in HDL (95% CI: -5.45-2.08, $p < 0.001$). The results are statistically significant at the alpha = 0.05 level.

QUESTION #2

The dataset `bweight.dta` contains information on 40 fictional births. The variables include birth weight in grams (`birthwt`), maternal age (`momage`) and gestational age of the infant at birth (`gestationalage`) in weeks. (Note that since this dataset is simulated, some of the values may fall outside realistic ranges.)

Question 2.1: Fit a regression for birth weight as a function of maternal and gestational age. Write a short description of the results

```
bweightLM <- lm(birthwt ~ gestationalage + momage, data = bweight)
```

```
> bweightLM <- lm(birthwt ~ gestationalage + momage, data = bweight)
> jtools::summ(bweightLM, confint = TRUE, digits = 3)
MODEL INFO:
Observations: 40
Dependent Variable: birthwt
Type: OLS linear regression

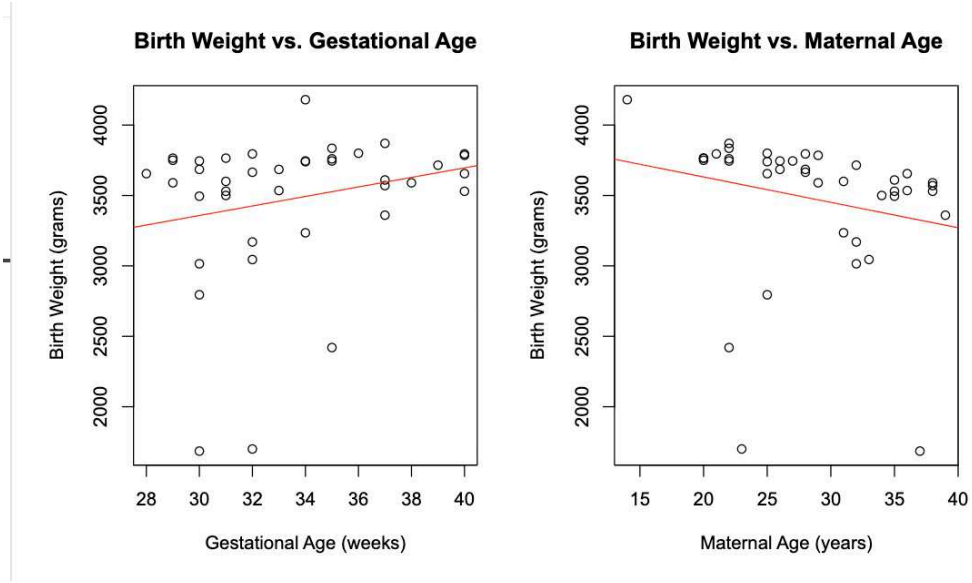
MODEL FIT:
F(2,37) = 3.063, p = 0.059
R² = 0.142
Adj. R² = 0.096

Standard errors:OLS
```

	Est.	2.5%	97.5%	t val.	p
(Intercept)	2623.082	1057.826	4188.337	3.396	0.002
gestationalage	47.294	-0.389	94.977	2.010	0.052
momage	-25.664	-52.090	0.763	-1.968	0.057

6.1

`Gestationalage`: The estimate shows that average infant birth weight increases 47.3 grams per week increase in gestational age (95% CI: -0.4 – 95.0, p = 0.052) and is significant at the alpha = 0.05 level.
`Momage`: The estimate also shows that infant birth weight is 25.7 grams lower per one year increase in maternal age (95% CI -52.1 -0.8, p = 0.0057) and is significantly significant at the alpha = 0.05 level.



Question 2.2: Check the assumption of normally distributed residuals in this dataset; does it appear reasonable?

```
# Create Residual Plot
residual_plot <- function(data) {
  g1 <- ggplot(data, aes(x = residuals)) +
    geom_histogram(color = "black", fill = "lightblue") +
    labs(title = "Histogram Plot of Residuals",
         x = "Residuals",
         y = "Count") +
    theme_minimal()

  g2 <- ggplot(data, aes(y = residuals)) +
    geom_boxplot(color = "black", fill = "lightblue") +
    labs(title = "Boxplot of Residuals",
         x = "",
         y = "Residuals") +
    theme_minimal()

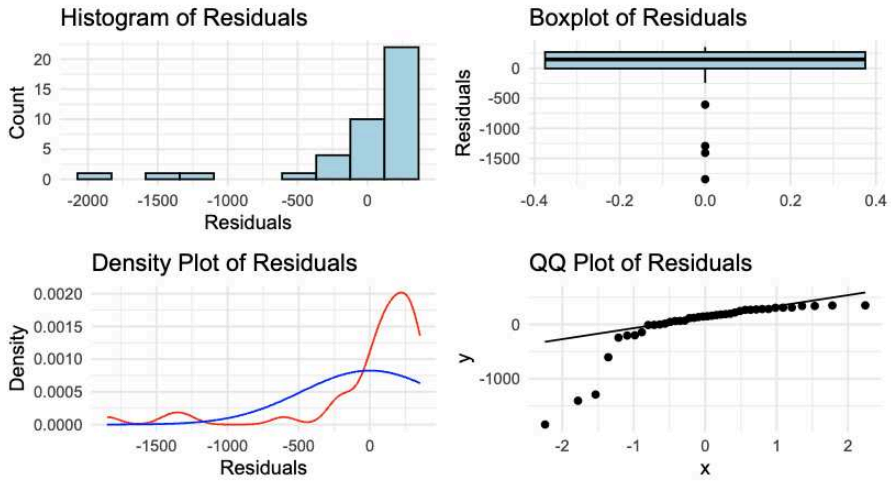
  g3 <- ggplot(data, aes(x = residuals)) +
    geom_density(color = "red") +
    stat_function(fun = dnorm,
                  args = list(mean = mean(data$residuals, na.rm = TRUE),
                              sd = sd(data$residuals, na.rm = TRUE)),
                  color = "blue") +
    labs(title = "Kernel Density Plot of Residuals",
         x = "Residuals",
         y = "Density") +
    theme_minimal()

  g4 <- ggplot(data, aes(sample = residuals)) +
    stat_qq() +
    stat_qq_line() +
    labs(title = "QQ Plot of Residuals") +
    theme_minimal()

  return(grid.arrange(g1, g2, g3, g4, nrow = 2))
}

# Apply the function to the bweight dataset
residual_plot(bweight)
```

7.1



The histogram, box plot, and QQ plots suggest a violation of the normality assumption. The tails of the distribution of the residuals are too long, as evident in the left tail, due to server outliers.

Question 2.3: Provide an alternate analysis that addresses violations of assumption of normality assessed in the previous question. Have your conclusions changed?

There are alternative tests that address violations of assumptions of normality. We can implement a Shapiro-Wilk test to test for departure from normality. From this, the Shapiro- Wilk test yielded a significant result ($W = 0.641$, $p < 0.001$) and we therefore reject the null hypothesis of normally distributed residuals. That said, statistical tests are not recommended, as rejection of normality for small violations can occur. The conclusions do not change, however, as graphical methods have demonstrated a violation of normality.

```
shapiro.test(bweight$residuals)
```

8.1

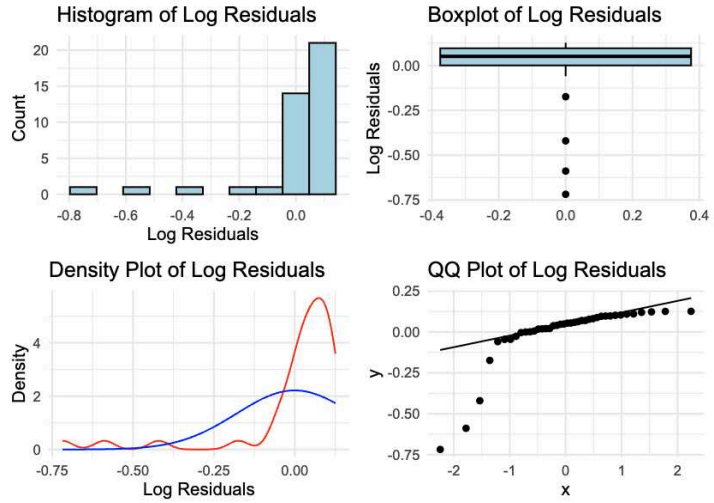
```
> shapiro.test(bweight$residuals)

        Shapiro-Wilk normality test

data:  bweight$residuals
W = 0.6407, p-value = 1.176e-08
```

In order to truly address violations of normality, we can log transform the outcome variable. The conclusions do not change, as the histogram, box plot, and QQ plots still suggest a violation of the normality assumption. It's important to note however, that the QQ plot is mostly linear, but deviates at the tails. In this case, without formally assessing outliers or the influence of individual data points, it is difficult to determine whether the departure from normality is due to truly non-normality of the residuals or the effect of a few extreme observations.

```
log_model <- lm(log(birthwt) ~ gestationalage + momage, data = bweight)
jtools::summ(log_model, confint = TRUE, digits = 3)
```



Question 2.4: Assess if there are any influential points and discuss their impact on the analysis results.

9.1

```
# Calculate sample size
n <- nrow(bweight)

# Calculate Cook's distance
cooks_d <- cooks.distance(bweightLM)
plot(cooks_d, type = "h", main = "Cook's Distance", ylab = "Cook's D", xlab = "Observation")
abline(h = 4/n, col = "red", lty = 2)  # Rule of thumb cutoff

# Identify influential points by Cook's distance
influential_cooks <- which(cooks_d > 4/n)
cat("Influential points by Cook's distance:", length(influential_cooks), "\n")
print(influential_cooks)

# Calculate DFBETAS
dfb <- data.frame(dfbetas(bweightLM))
ref_line <- 2/sqrt(n)  # Standard cutoff for dfbetas

# Create data frame for plotting DFBETAS
dfb_data <- data.frame(
  Variable = rep(c(
    "Intercept",
    "Gestational Age",
    "Maternal Age"
  ), each = nrow(dfb)),
  DFBETA = c(
    dfb$X.Intercept.,
    dfb$gestationalage,
    dfb$momage
  ),
  Observation = rep(1:nrow(dfb), 3)
)
```

```
# Plot DFBETAS
ggplot(dfb_data, aes(x = Observation, y = DFBETA, color = Variable)) +
  geom_point() +
  geom_hline(yintercept = c(ref_line, -ref_line), linetype = "dashed") +
  facet_wrap(~Variable, scales = "free_y") +
  labs(title = "DFBETAS for Birth Weight Model",
       x = "Observation",
       y = "DFBETA Value") +
  theme_minimal()

# Identify influential points by DFBETAS
gestage_influential <- which(abs(dfb$gestationalage) > ref_line)
momage_influential <- which(abs(dfb$momage) > ref_line)
all_influential <- unique(c(gestage_influential, momage_influential))

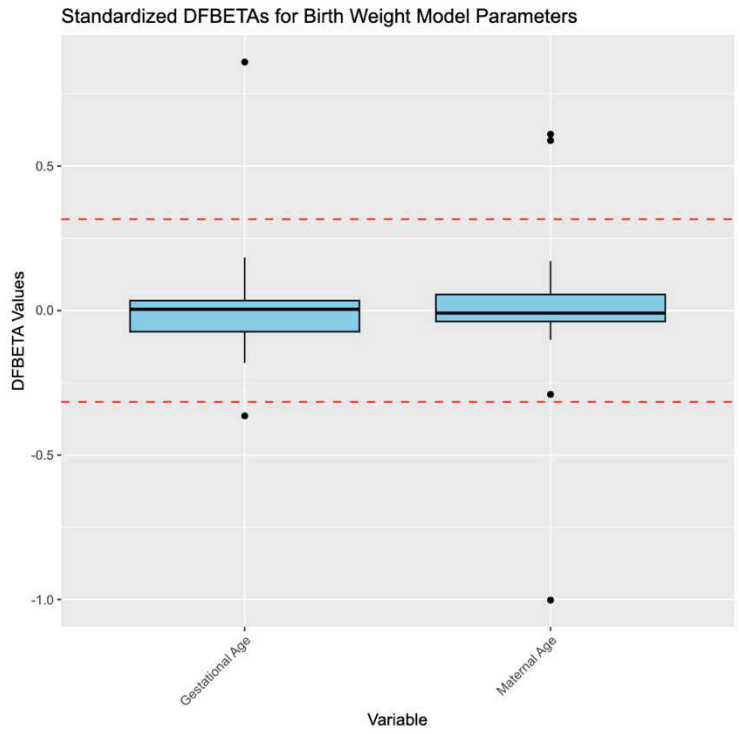
cat("\nInfluential points for gestational age:", length(gestage_influential), "\n")
print(gestage_influential)

cat("\nInfluential points for maternal age:", length(momage_influential), "\n")
print(momage_influential)

cat("\nTotal unique influential points:", length(all_influential), "\n")
print(all_influential)

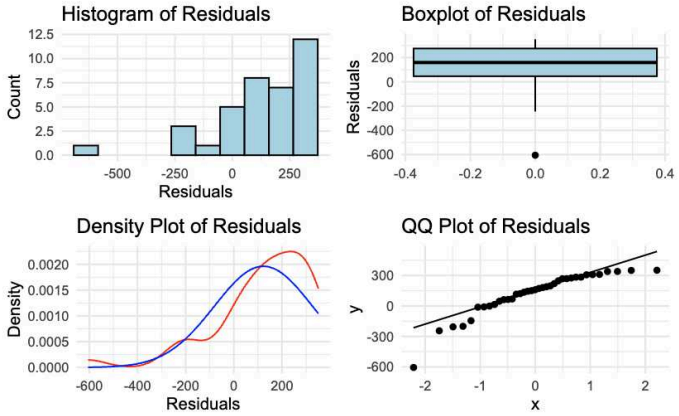
# Look at the values of influential points
if(length(all_influential) > 0) {
  cat("\nValues of influential observations:\n")
  print(bweight[all_influential, c("birthwt", "gestationalage", "momage")])
}

# Fit model without influential points
if(length(all_influential) > 0) {
  bweightLM_no_influential <- lm(birthwt ~ gestationalage + momage,
                                 data = bweight[-all_influential, ])
```



Boxplots of the DFBETA statistics for gestational age and maternal age identified a small set of extreme values, suggesting potential influence on the fitted regression model. As an ad hoc sensitivity analysis, we omitted these potentially (3) influential observations and refit the model. Before removal, the gestational age coefficient was 47.294 and marginally significant at the alpha = 0.05 level (p = 0.052), but after removal, the coefficient was 33.617 and statistically significant at the alpha = 0.01 level (p = 0.002). The maternal age coefficient shifted from -25.664 (p = 0.0057) to -25.980 (p = 0.00), showing a stronger statistical significance. Lastly, the plotted residuals without the 3 points show far less deviation from normality.

```
# Identify influential points by DFBETAS
gestage_influential <- which(abs(dfb$gestationalage) > ref_line)
momage_influential <- which(abs(dfb$momage) > ref_line)
all_influential <- unique(c(gestage_influential, momage_influential))
# Fit model without influential points
if(length(all_influential) > 0) {
  bweightLM_no_influential <- lm(birthwt ~ gestationalage + momage,
                                 data = bweight[-all_influential, ])
```




```
> bweightLM <- lm(birthwt ~ gestationalage + momage, data = bweight)
> jtools::summ(bweightLM, confint = TRUE, digits = 3)
MODEL INFO:
Observations: 40
Dependent Variable: birthwt
Type: OLS linear regression

MODEL FIT:
F(2,37) = 3.063, p = 0.059
R² = 0.142
Adj. R² = 0.096

Standard errors:OLS
```

	Est.	2.5%	97.5%	t val.	p
(Intercept)	2623.082	1057.826	4188.337	3.396	0.002
gestationalage	47.294	-0.389	94.977	2.010	0.052
momage	-25.664	-52.090	0.763	-1.968	0.057

```
> jtools::summ(bweightLM_no_influential, confint = TRUE, digits = 3)
MODEL INFO:
Observations: 36
Dependent Variable: birthwt
Type: OLS linear regression

MODEL FIT:
F(2,33) = 10.183, p = 0.000
R² = 0.382
Adj. R² = 0.344

Standard errors:OLS
```

	Est.	2.5%	97.5%	t val.	p
(Intercept)	3210.105	2555.160	3865.050	9.972	0.000
gestationalage	33.617	12.899	54.335	3.301	0.002
momage	−25.980	−38.832	−13.127	−4.113	0.000

QUESTION #3

This question is based on a subset of variables from the Western Collaborative Group Study (WCGS) data. The dataset `final-wcgs.dta` includes the following variables measured at the enrollment visit: `dibpat`, a binary indicator of a type-A behavior (with type-B as the reference category); `smoke`, a binary indicator of reporting smoking at least one cigarette per day; `bmi` (body mass index); `chol`, total serum cholesterol in mg/dl; `age` (years), `sbp`, systolic blood pressure (mmHg); `ncigs` (number of cigarettes smoked per day), and `npacks`, a categorical indicator based on `ncigs` of number of packs smoked per day (with category levels corresponding to 0, < ½, ½-1, >1 – 2, >2 packs per day). The outcome variable `chd69` is a binary indicator of a coronary heart disease (CHD)-related event occurring during study follow-up.

The goal is to investigate smoking, as measured by the binary indicator `smoke`, as a risk factor for CHD, given the available predictor variables. For questions 3.1 - 3.3, *focus only on this binary measure* as the predictor of primary interest - you will investigate quantitative measures of smoking (`ncigs` and `npacks`) in question 4.

Question 3.1: Write a sentence summarizing the crude (or unadjusted) effect of smoking on CHD, using the binary indicator `smoke` of baseline smoking status.

```
model <- glm(chd69 ~ smoke, family = binomial, data = wcgs)
jtools::summ(model, confint = TRUE, digits = 3, exp = TRUE)

# Unadjusted odds ratio for effect of smoking on CHD
oddsratio(wcgs$chd69, wcgs$smoke)

# Alternative approach using epitools
epitab(wcgs$smoke, wcgs$chd69, method = "oddsratio")
```

13.1

```
> model <- glm(chd69 ~ smoke, family = binomial, data = wcgs)
> jtools::summ(model, confint = TRUE, digits = 3, exp = TRUE)
MODEL INFO:
Observations: 3154
Dependent Variable: chd69
Type: Generalized linear model
  Family: binomial
  Link function: logit

MODEL FIT:
χ²(1) = 22.863, p = 0.000
Pseudo-R² (Cragg-Uhler) = 0.017
Pseudo-R² (McFadden) = 0.013
AIC = 1762.381, BIC = 1774.494

Standard errors:MLE
------------------------------------------------------------
                   exp(Est.)    2.5%   97.5%    z val.       p
---------------- ----------- ------- ------- --------- -------
(Intercept)            0.063   0.051   0.077   -26.535   0.000
smokesmoker            1.877   1.445   2.440     4.710   0.000
------------------------------------------------------------
> □
```

```
> # Unadjusted odds ratio for effect of smoking on (
> oddsratio(wcgs$chd69, wcgs$smoke)
$data
         Outcome
Predictor nonsmoker smoker Total
    no         1554   1343  2897
    yes          98    159   257
    Total      1652   1502  3154

$measure
         odds ratio with 95% C.I.
Predictor estimate    lower    upper
      no  1.000000       NA       NA
      yes 1.875857 1.445907 2.445018

$p.value
         two-sided
Predictor   midp.exact fisher.exact   chi.square
      no            NA           NA           NA
      yes 1.834284e-06 2.365647e-06 1.831404e-06
```

Participants from the WCGS study who report smoking at least one cigarette per day have an 87% increased odds of coronary heart disease related events as compared with adults who report not smoking (OR :1.87, p < 0.001).

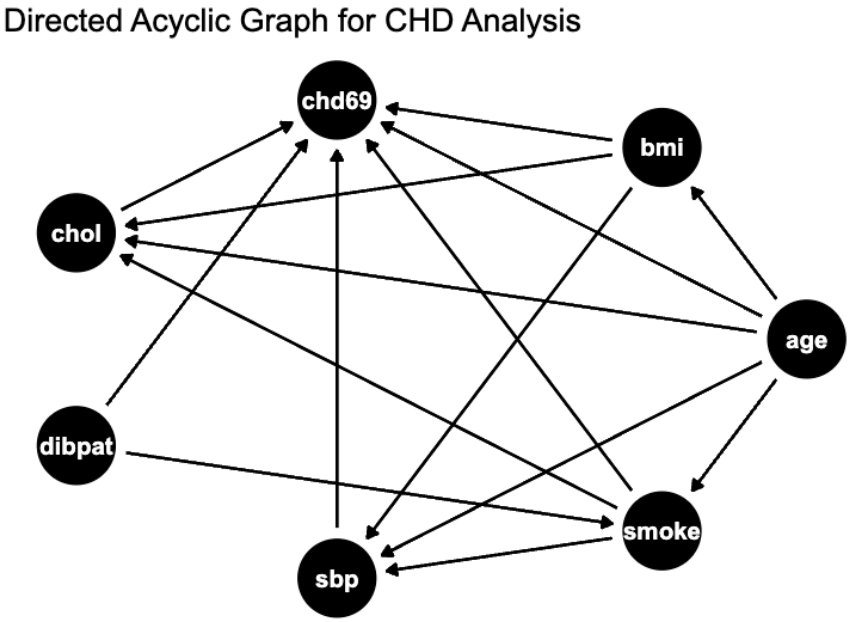
Question 3.2: Fit a logistic or log relative risk regression model to estimate the effect smoking on CHD adjusted for the possible confounding influence of other enrollment variables. Focus again on the `smoke` indicator as the predictor of primary interest, and on providing an estimate that addresses the possible causal effect of baseline smoking status on CHD risk. Assume also that any included non-categorical predictors can be adequately modeled by linear terms, and that interactions can be ignored. Answer the following questions:

(i) Describe how you selected adjustment variables for your model including a discussion of what you assumed about confounding or mediating roles of the variables considered. Also, identify your chosen measure of the effect of smoking. Note that a conditional estimate (e.g. adjusted regression coefficient), or a marginal estimate (e.g. a marginal odds ratio or relative risk) is acceptable as an effect measure.

Adjustment variables were selected by first drawing a DAG which reflects present causal relationships between variables on subject matter knowledge. 14.1

1. **Personality type:** Type A behavior has been linked to increased risk in CHD. Personality type is also linked to smoking behavior.
2. **Age:** Age affects CHD risk and smoking habits, age also has relationships with bmi, sbp, and chol
3. **BMI, CHOL, SBP** are also known risk factors of CHD
4. **BMI**: linked to sbp, chol.

From this, we assume the effect of smoking on CHD is confounded by age and personality type. From other subject matter knowledge, we know that blood pressure (sbp) is increased by smoking, which in turn increases CHD risk. We also find that smoking affects cholesterol levels(chol), which also in turn increase CHD risk. Therefore, we find that cholesterol and blood pressure are mediators on the causal pathway between smoking and CHD risk and should therefore not be adjusted in the model. We also don't adjust for bmi because, smoking influences bmi, and not the other way around. From this analysis and background on subject matter knowledge, we therefore only adjust for age and dibpat. As we are implementing logistic regression, we will use the adjusted odds ratio. Important to note, CHD is a rare outcome in this data set (257 events; 8% of observations) and the odds ratio can therefore be approximately equal to the relative risk.



(ii) Present the estimated adjusted effect measure with associated 95% confidence intervals.

a. The adjusted odds ratio for smokers compared to nonsmokers is 1.825 (95% CI 1.40 – 2.38, p < 0.001) after controlling for age and behavior pattern.

b. The adjusted odds ratio for Type A behavior compared to type B behavior is 2.16 (95% CI 1.64 – 2.85, P < 0.001)

c. The adjusted odds ratio for age was 1.07 (95% CI 1.05 – 1.10, P < 0.001) for each one-year increase in age, after adjusting for smoking status and behavior pattern.

```
model_adjust <- glm(chd69 ~ smoke + age + dibpat,
                    family = binomial,
                    data = wcgs)
tools::summ(model_adjust, confint = TRUE, digits = 3, exp = TRUE)
```

```
> jtools::summ(model_adjust, confint = TRUE, digits = 3, exp = TRUE)
MODEL INFO:
Observations: 3154
Dependent Variable: chd69
Type: Generalized linear model
  Family: binomial
  Link function: logit

MODEL FIT:
χ²(3) = 97.348, p = 0.000
Pseudo-R² (Cragg-Uhler) = 0.070
Pseudo-R² (McFadden) = 0.055
AIC = 1691.895, BIC = 1716.121

Standard errors:MLE
------------------------------------------------------------------
                   exp(Est.)    2.5%   97.5%    z val.        p
----------------- ---------- ------- ------- --------- -------
(Intercept)            0.001   0.000   0.005   -11.480   0.000
smokesmoker            1.825   1.399   2.381     4.440   0.000
age                    1.072   1.049   1.097     6.092   0.000
dibpatA1,A2            2.158   1.635   2.850     5.426   0.000
------------------------------------------------------------------
```

15.1

(iii) Provide a brief paragraph summarizing your findings and describe issues with and/or limitations of your analysis.

In our analysis, we found that smoking at least one cigarette a day is significantly associated with an increased odds of CHD events, with an adjusted odds ratio of 1.83 (94% CI 1.40 – 2.38, p <0.001), after controlling for age and behavior pattern. A likelihood ratio test comparing unadjusted and adjusted models also showed significant improvement in model fit when including age and behavioral pattern ($p < 0.001$). The bar plots below illustrate the predicted probabilities of CHD under both the unadjusted and adjusted logistic regression models across different combinations of smoking status, age, and behavior type where probabilities are calculated using the equations provided in the descriptions of the figures. Across all visualized subgroups, smoking consistently increases the predicted probability of CHD, but in the adjusted model, the magnitude of increase varies by age and behavior type demonstrating plausible evidence of confounding by age and behavior. Limitations of this analysis include lack of interaction terms between smoking and other predictors, like age or behavior type. Additionally, representation of CHD and smoking status as binary predictors and outcomes can obscure meaningful differences if the predictors or outcomes have severity levels, so collapsing these variables may result in a loss of information. Consider that the predictor for smoking is just smoke one cigarette per day and therefore does not capture the nuance between a 5 pack a day smoker and a 1 cigarette per day smoker.

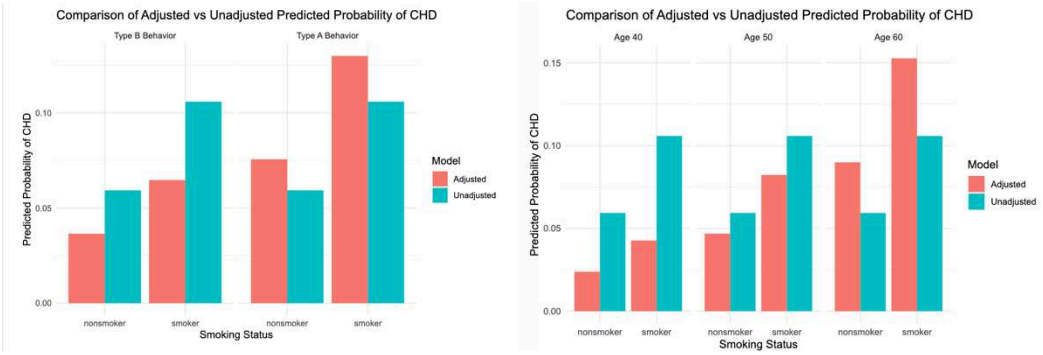


Figure 1. Behavior Pattern Comparison

a. **Adjusted model (red)**: $P(CHD = 1) = \exp(\beta_0 + \beta_1 \times smoke + \beta_2 \times age + \beta_3 \times dibpat) / [1 + \exp(\beta_0 + \beta_1 \times smoke + \beta_2 \times age + \beta_3 \times dibpat)]$
   - age is fixed at the mean value

b. **Unadjusted model (blue)**: $P(CHD = 1) = \exp(\beta_0 + \beta_1 \times smoke) / [1 + \exp(\beta_0 + \beta_1 \times smoke)]$

Figure 2. Age Comparison:

- Predicted probabilities by smoking status and age
  a. **Adjusted model (red)**: $P(CHD = 1) = \exp(\beta_0 + \beta_1 \times smoke + \beta_2 \times age + \beta_3 \times dibpat) / [1 + \exp(\beta_0 + \beta_1 \times smoke + \beta_2 \times age + \beta_3 \times dibpat)]$
     - dibpat is fixed at "B3,B4" (Type B behavior)
  b. **Unadjusted model (blue)**: $P(CHD = 1) = \exp(\beta_0 + \beta_1 \times smoke) / [1 + \exp(\beta_0 + \beta_1 \times smoke)]$

Question 3.3: Fit an additional regression model including any variables adjusted for in the previous analysis, and also including a two-way interaction between the binary indicators `smoke` and `dibpat`.

```
# Fit model with interaction between smoke and dibpat
model_interaction <- glm(chd69 ~ smoke * dibpat + age,
                         family = binomial,
                         data = wcgs)

# Display model summary with odds ratios
jtools::summ(model_interaction, confint = TRUE, digits = 3, exp = TRUE)
```

```
> jtools::summ(model_interaction, confint = TRUE, digits = 3, exp = TRUE)
MODEL INFO:
Observations: 3154
Dependent Variable: chd69
Type: Generalized linear model
  Family: binomial
  Link function: logit

MODEL FIT:
χ²(4) = 99.117, p = 0.000
Pseudo-R² (Cragg-Uhler) = 0.072
Pseudo-R² (McFadden) = 0.056
AIC = 1692.126, BIC = 1722.408

Standard errors:MLE
------------------------------------------------------------------------------
                              exp(Est.)    2.5%   97.5%    z val.        p
------------------------------ ---------- ------- ------- --------- -------
(Intercept)                        0.001   0.000   0.004   -11.482   0.000
smokesmoker                        2.370   1.476   3.808     3.569   0.000
dibpatA1,A2                        2.729   1.736   4.288     4.353   0.000
age                                1.072   1.048   1.096     6.074   0.000
smokesmoker:dibpatA1,A2            0.680   0.383   1.205    -1.322   0.186
------------------------------------------------------------------------------
```

17.1

(i) Summarize evidence for (multiplicative) interaction from your fitted model.

a. **Smokessmoker**: Among participants with Type B behavior (dibpat = 0), smoking was associated with more than a twofold increase odds of coronary heart disease (OR = 2.370, 95% CI: 1.48 – 3.81, p < 0.001)

b. **DibpatA1, A2:** Among nonsmokers, those with type A behavior had significantly higher odds of CHD compared to those with type B behavior (OR = 2.729, 95% CI: 1.74-4.29, p < 0.001)

c. **Smokessmoker:dibpatA1,A2:** The interaction term between smoking and Type A behavior has an odds ratio of 0.680(95% CI: 0.38-1.21, p = 0.186) and is not statistically significant at the 0.005 level (z = -1.322, p = 0.186).

(ii) Present the estimated associations between the `smoke` variable and CHD for both type A and B behavior categories (including 95% confidence intervals)

a. Among individuals with Type B behavior, smoking was associated with an 135% increased odds of coronary heart disease (CHD) compared to non-smokers (OR: 2.35, 95% CI: 1.47-3.77, p < 0.001)

17.2

b. Among participants with type A behavior, smokers had 60% increased odds of CHD compared to nonsmokers (OR: 1.60, 95% CI: 1.17-2.19, p = 0.004)

```
> # Calculate average marginal effects of smoking within each behavior pattern
> smoke_effects <- avg_comparisons(
+   model_interaction,
+   variables = "smoke",
+   by = "dibpat",
+   comparison = "lnor",  # log odds ratio
+   transform = exp       # exponentiate to get odds ratios
+ )
> # Print the results
> print(smoke_effects)

 dibpat Estimate Pr(>|z|)    S 2.5 % 97.5 %
  B3,B4     2.35  < 0.001 11.4  1.47   3.77
  A1,A2     1.60  0.00362  8.1  1.17   2.19
```

(iii) Provide a summary of evidence additive interaction between `smoke` and `dibpat`, using the relative excess risk for interaction (*RERI*) measure, and give the accompanying 95% confidence interval and *p*-value. (Note that if you used logistic regression, you may assume that the outcome is sufficiently rare that the odds ratio-based version of *RERI* is acceptable.) 18.1

a. The estimated relative risk due to interaction (RERI) was 0.297 (95% CI -1.04 – 1.64, p = 0.665) indicating positive interaction but, no statistically significant departure from additivity. Also, the 95% confidence interval includes the null value, indicating insufficient evidence for additive interaction.

```
# Calculate RERI (Relative Excess Risk due to Interaction) using hypotheses function
hypotheses(model_interaction,
           "exp(smokesmoker + `dibpatA1,A2` + `smokesmoker:dibpatA1,A2`) -
exp(smokesmoker) - exp(`dibpatA1,A2`) + 1 = 0")

                                                                                Hypothesis
  exp(smokesmoker+`dibpatA1,A2`+`smokesmoker:dibpatA1,A2`)−exp(smokesmoker)−exp(`dibpatA1,A2`)+1=0
  Estimate Std. Error     z Pr(>|z|)   S 2.5 % 97.5 %
     0.297      0.684 0.434    0.665 0.6 -1.04   1.64
```

This question is based on the Western Collaborative Group Study (WCGS) dataset `final-wcgs.dta` considered in the previous question and focuses on a subsidiary analysis of the relationship between smoking and CHD risk based on the quantitative variable `ncigs` (number of cigarettes smoked per day) and the categorical indicator `npacks` of number of packs smoked per day.

Question 4.1: Fit a logistic regression model for the relationship between the CHD outcome `chd69` and `ncigs`, including `age` (as a linear term) and behavior pattern (`dibpat`) as adjustment variables.

```
fit_linear <- glm(chd69 ~ ncigs + age + dibpat, data = wcgs, family =
"binomial")
jtools::summ(fit_linear, confint = TRUE, digits = 3, exp = TRUE)
```

```
Odds ratio per cigarette: 1.022 95% CI: [ 1.014 - 1.03 ]> # Fit logistic regression model
> fit_linear <- glm(chd69 ~ ncigs + age + dibpat, data = wcgs, family = "binomial")
> jtools::summ(fit_linear, confint = TRUE, digits = 3, exp = TRUE)
MODEL INFO:
Observations: 3154
Dependent Variable: chd69
Type: Generalized linear model
  Family: binomial
  Link function: logit

MODEL FIT:
χ²(3) = 103.964, p = 0.000
Pseudo-R² (Cragg-Uhler) = 0.075
Pseudo-R² (McFadden) = 0.058
AIC = 1685.279, BIC = 1709.505

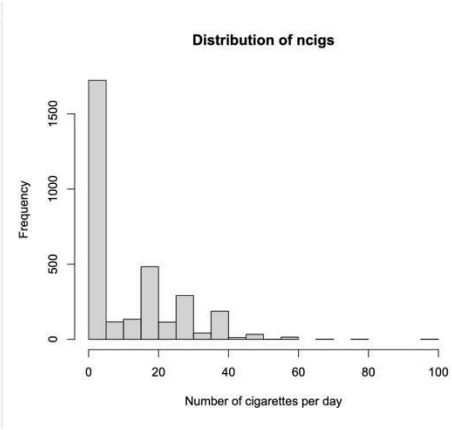
Standard errors:MLE
------------------------------------------------------------------
                    exp(Est.)    2.5%   97.5%    z val.       p
------------------ ----------- ------- ------- --------- -------
(Intercept)             0.001   0.000   0.004   -11.501   0.000
ncigs                   1.022   1.014   1.030     5.295   0.000
age                     1.074   1.050   1.098     6.192   0.000
dibpatA1,A2             2.094   1.584   2.767     5.191   0.000
------------------------------------------------------------------
```

(i) Summarize evidence for a linear association between the log odds of CHD and `ncigs`.

We find for each additional cigarette smoked per day, the odds of CHD increase by 2.2% and are statistically significant at the alpha = 0.001 level (z = 5.295, p = 0.001).

(ii) Discuss any limitations of the linear model fitted in part (*i*). (Consider the relative frequency of observed values of `ncigs` in your answer.)

When examining the distribution of ncigs in the barplot below, we see that the number of cigs per day is highly skewed, with the majority of observations clustered at low values. This inturn reduces the precision of coefficeint estimaes, which depend of the variance of the predictor. With so few data points at high values of ncigs, we likely observe wider confidence intervals and lower precision.



(iii) Conduct additional analyses to evaluate the plausibility of a nonlinear association between `ncigs` and the log odds of CHD, and write a brief summary of your findings.

In order to test for departure from linearity, a restricted cubic spline transformation of ncigs was constructed using four knots placed at values calculated based on relative frequencies and distributions of smoking quantities. About 52% are nonsmokers, with peaks at 10 cigarettes (2.6%), 15 cigarettes(3.2%), 20 cigarettes(14.4%), 25 cigarettes(3.3%), 30 cigarettes(9%), and 40 cigarettes(5.8%). We create knots for a restricted cubic spline with the following values: (10,20,20,40) corresponding to a half pack, one pack, one and a half packs, and two packs.

Our resulting spline model was fit using logistic regression, adjusting for age and behavioral pattern and the table output shows both spline terms are statistically significant: rcspline1:0.150 (95% CI: 0.084 – 0.216, z = 4.467, p < 0.001) and rcspline2: -0.480(95% CI: -0.718 - - 0.241, z = -3.945, p < 0.001). However, we focus on the second spline to suggest a statistically significant departure from linearity and evidence of a nonlinear association between the number of cigarettes smoked per day and the log odds of coronary heart disease (CHD). The figure below demonstrates the adjusted relationship between the number of cigarettes smoked per day and the log odds of CHD using the restricted cubic spline.

Lastly, we conduct a likelihood ratio test comparing the linear model to the spline model and find a deviance difference of -1.61 (p = 0.205) indicating there is not a statistically significant difference

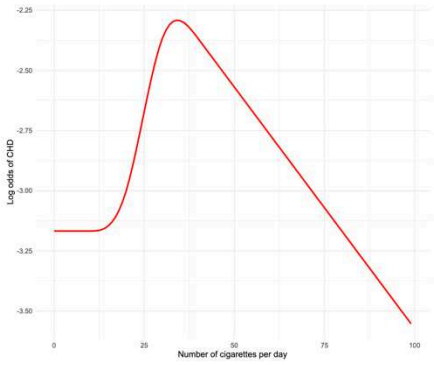
between the two models. So, there is a statistically significant departure from linearity, but the nonlinear model does not have a statistically significant better fit than the linear mode.

MODEL INFO:
Observations: 3154
Dependent Variable: chd69
Type: Generalized linear model
Family: binomial
Link function: logit

MODEL FIT:
$\chi^2(4) = 102.358$, $p = 0.000$
Pseudo-R^2 (Cragg-Uhler) = 0.074
Pseudo-R^2 (McFadden) = 0.057
AIC = 1688.886, *BIC* = 1719.168

Standard errors:MLE

	Est.	2.5%	97.5%	z val.	p
(Intercept)	-6.502	-7.613	-5.391	-11.469	0.000
rcspline1	0.150	0.084	0.216	4.467	0.000
rcspline2	-0.480	-0.718	-0.241	-3.945	0.000
age	0.072	0.049	0.095	6.258	0.000
dibpatA1,A2	0.751	0.472	1.030	5.275	0.000




```
Model 1: chd69 ~ ncigs + age + dibpat
Model 2: chd69 ~ rcspline + age + dibpat
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      3150     1677.3
2 _    3149     1678.9  1  -1.6065
```

```
# Create restricted cubic spline for ncigs
# Define knots with meaningful smoking quantities
knots <- c(10, 20, 30, 40)
wcgs$rcspline <- rcspline.eval(wcgs$ncigs, knots = knots)

# Fit model with splines
fit_nonlinear <- glm(chd69 ~ rcspline + age + dibpat, data = wcgs, family =
"binomial")
jtools::summ(fit_nonlinear, confint = TRUE, digits = 3)

# Test for nonlinearity (joint test of all spline terms except the first)
anova(fit_linear, fit_nonlinear, test = "Chisq")

# Visualize the relationship
new_data <- data.frame(
  ncigs = seq(0, max(wcgs$ncigs), length.out = 100),
  age = mean(wcgs$age),
  dibpat = "B3,B4"  # Using Type B behavior as reference
)
new_data$rcspline <- rcspline.eval(new_data$ncigs, knots = knots)
new_data$predicted_log_odds <- predict(fit_nonlinear, newdata = new_data)

# Plot the relationship
ggplot(new_data, aes(x = ncigs, y = predicted_log_odds)) +
  geom_line(color = "red", linewidth = 1) +
  labs(x = "Number of cigarettes per day", y = "Log odds of CHD") +
  theme_minimal()
```

Question 4.2: Fit a logistic regression model for the relationship between the CHD outcome `chd69` and the categorical variable `npacks`, again including `age` and `dibpat` as adjustment variables.

(i) Summarize the overall effect of smoking (as measured by `npacks`) on the odds of CHD using a likelihood ratio test.

The likelihood ratio test compared the model with npacks(smoking) to one without, while adjusting for age and behavior pattern. The test was statistically significant ($X^2 = 28.18$, df = 4, $p < 0.001$) indicating that smoking, as categorized by number of packs per day, has a statistically significant overall effect on the odds of CHD.

```
# Question 4.2 (i): Fit full model with npacks
fit_full <- glm(chd69 ~ npacks + age + dibpat, data = wcgs, family = "binomial")
jtools::summ(fit_full, confint = TRUE, digits = 3, exp = TRUE)

# Fit reduced model without npacks
fit_reduced <- glm(chd69 ~ age + dibpat, data = wcgs, family = "binomial")
jtools::summ(fit_reduced, confint = TRUE, digits = 3, exp = TRUE)

# Likelihood ratio test
anova(fit_reduced, fit_full, test = "Chisq")
```

```
Model 1: chd69 ~ age + dibpat
Model 2: chd69 ~ npacks + age + dibpat
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1      3151     1704.2
2      3147     1676.0  4   28.182 1.146e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

(ii) Perform a test for linear trend of the association between and the log odds of CHD and `npacks`, and summarize the results.

We implemented four tests for a linear trend in the association between smoking (npacks) and CHD. The first, a chi squared test for trend in proportions showed strong evidence of a linear increase in CHD risk with increasing npacks ($X^2 = 32.3$, df = 1, $p < 0.001$). A logistic regression model treating npacks as continuous found similar results (($\beta = 0.259$, 95% CI: 0.162–0.355, $p < 0.001$). Our likelihood ratio test comparing models found no evidence of departure from linearity ($p = 0.89$). Finally, a test for contrasts found the linear contrast for npacks was statistically significant ($\beta = 0.947$, $p < 0.001$).

```
> # Approach 1: Using test for trend for binary outcome
> # First create a contingency table of CHD by npacks
> tab <- table(wcgs$chd69, wcgs$npacks)
> # Calculate column percentages
> col_percentages <- prop.table(tab, margin = 2) * 100
> print(addmargins(col_percentages)) # proportions

             0       <1/2      1/2-1       >1-2         >2        Sum
  no   94.067797  93.333333  91.000000  87.854890  80.952381 447.208400
  yes   5.932203   6.666667   9.000000  12.145110  19.047619  52.791600
  Sum 100.000000 100.000000 100.000000 100.000000 100.000000 500.000000
> # Test for trend using prop.trend.test
> # Extract the counts of CHD for each npacks level
> chd_counts <- tab["yes", ]
> # Get the total counts for each npacks level
> total_counts <- colSums(tab)
> # Run the trend test
> prop.trend.test(chd_counts, total_counts)

        Chi-squared Test for Trend in Proportions

data:  chd_counts out of total_counts ,
 using scores: 1 2 3 4 5
X-squared = 32.226, df = 1, p-value = 1.373e-08

> # Approach 2: Using logistic regression to test for linear trend
> # Create a model with npacks as a continuous variable
> # First check if npacks is a factor
> class(wcgs$npacks)
[1] "factor"
> # Create a numeric score for npacks if needed
> wcgs$npacks_score <- as.numeric(wcgs$npacks)
> # Test for trend by treating npacks as a continuous predictor
> fit_trend <- glm(chd69 ~ npacks_score + age + dibpat, data = wcgs, family = "binomial")
> jtools::summ(fit_trend, confint = TRUE, digits = 3)
MODEL INFO:
Observations: 3154
Dependent Variable: chd69
Type: Generalized linear model
  Family: binomial
  Link function: logit


MODEL FIT:
χ²(3) = 104.639, p = 0.000
Pseudo-R² (Cragg-Uhler) = 0.076
Pseudo-R² (McFadden) = 0.059
AIC = 1684.604, BIC = 1708.830

Standard errors:MLE
------------------------------------------------------------
                      Est.     2.5%    97.5%    z val.       p
------------------ -------- -------- -------- --------- -------
(Intercept)          -6.833   -7.970   -5.697   -11.782   0.000
npacks_score          0.259    0.162    0.355     5.249   0.000
age                   0.071    0.048    0.094     6.171   0.000
dibpatA1,A2           0.748    0.470    1.027     5.264   0.000
------------------------------------------------------------
> # Test for departure from linearity by comparing models
> # This compares the model with npacks as continuous vs categorical
> fit_cat <- glm(chd69 ~ factor(npacks) + age + dibpat, data = wcgs, family = "binomial")
> anova(fit_trend, fit_cat, test = "Chisq")
Analysis of Deviance Table

Model 1: chd69 ~ npacks_score + age + dibpat
Model 2: chd69 ~ factor(npacks) + age + dibpat
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      3150     1676.6
2      3147     1676.0  3  0.62611   0.8904
```

```
------------------------------------------------------------------
                      Est.     2.5%    97.5%    z val.       p
------------------ -------- -------- -------- --------- -------
(Intercept)          -6.074   -7.191   -4.957   -10.661   0.000
npacks_ord.L          0.947    0.448    1.447     3.718   0.000
npacks_ord.Q          0.160   -0.284    0.604     0.706   0.480
npacks_ord.C         -0.076   -0.633    0.480    -0.268   0.788
npacks_ord^4          0.048   -0.391    0.487     0.215   0.830
age                   0.071    0.049    0.094     6.183   0.000
dibpatA1,A2           0.743    0.464    1.023     5.218   0.000
------------------------------------------------------------------
```

```
# Approach 1: Using test for trend for binary outcome
# First create a contingency table of CHD by npacks
tab <- table(wcgs$chd69, wcgs$npacks)
# Calculate column percentages
col_percentages <- prop.table(tab, margin = 2) * 100
print(addmargins(col_percentages)) # proportions

# Test for trend using prop.trend.test
# Extract the counts of CHD for each npacks level
chd_counts <- tab["yes", ]
# Get the total counts for each npacks level
total_counts <- colSums(tab)
# Run the trend test
prop.trend.test(chd_counts, total_counts)

# Approach 2: Using logistic regression to test for linear trend
# Create a model with npacks as a continuous variable
# First check if npacks is a factor
class(wcgs$npacks)
```

```
# Create a numeric score for npacks if needed
wcgs$npacks_score <- as.numeric(wcgs$npacks)
# Test for trend by treating npacks as a continuous predictor
fit_trend <- glm(chd69 ~ npacks_score + age + dibpat, data = wcgs, family = "binomial")
jtools::summ(fit_trend, confint = TRUE, digits = 3)

# Test for departure from linearity by comparing models
# This compares the model with npacks as continuous vs categorical
fit_cat <- glm(chd69 ~ factor(npacks) + age + dibpat, data = wcgs, family = "binomial")
anova(fit_trend, fit_cat, test = "Chisq")

# Test for linearity
# Set polynomial contrasts for npacks
wcgs$npacks_ord <- wcgs$npacks  # make a copy to preserve factor levels
contrasts(wcgs$npacks_ord) <- contr.poly(nlevels(wcgs$npacks_ord))

# Fit logistic regression with contrasts
fit_contrast <- glm(chd69 ~ npacks_ord + age + dibpat, data = wcgs, family = "binomial")
jtools::summ(fit_contrast, confint = TRUE, digits = 3)
```

(iii) Discuss briefly the limitations/ merits of the analysis conducted in question 4.2 compared to the analysis conducted in question 4.1. You may also suggest alternative analyses.

Question 4.2 relies on ordered categories rather than continuous and while simpler in interpretation, there may be nuance in the continuous variable that the categorical might not detect. 4.1 also allows greater interpretation and visitation in departure from linearity. One approach would be to apply cubic splines to transformed scores of npacks. Another approach would be to transform ncigs into npacks directly by binning the continuous variables into ordinal pack based categories, unless that is already how the variable itself was created.

Final

Anya DeCarlo

2025-03-15

Question 1

QUESTION #1

The dataset final-hersdata.dta is a subset of data from the HERS trial.

Focus on the outcome variable HDL (high-density lipoprotein in mg/dL), and the following three predictors: physical activity physact, age and body mass index BMI. The variable physact is coded 1: much less active, 2: somewhat less active (labeled as “less” in the dataset), 3: about as active (labeled as “equally” in the dataset), 4: somewhat more active, 5: much more active.

Question 1.1: (i) How would you classify the data type of the predictor physact? (ii) Is a test for linear trend appropriate for this predictor? Why? (iii) Fit a regression model with HDL as the outcome and physact as a predictor. (iv) Perform a test for linear trend. (Evaluating departure from linear trend is optional.) (v) Write 2-4 sentences describing, as you would in a paper, the estimated effect of physical activity on HDL levels. (Include 95% CIs when appropriate!)

Question 1.2: (i) Assume that it is reasonable to consider age and BMI as possible confounding influences on the relationship between physical activity and HDD. Adjust the relationship in Q1.1 for the variables age and BMI by including them as predictors in the regression model. (You may assume that these variables can be included as linear terms.) Interpret the results, and answer the following question: (ii) Are the results consistent with age and BMI having a confounding influence on the relationship between physact and HDL? (iii) Write 2-4 sentences describing the adjusted association between physact and HDL.

Question 1.3: (i) Based on the model in Q1.2, estimate the mean difference in HDL between a person who is ”much more” active compared with someone who is ”about as” active (adjusting for age and BMI). (ii) Give the 95% confidence interval for this effect

Question 1.4: (i) Based on the model in Q1.2, estimate the effect on mean HDL associated with a 10-year decrease in age (adjusting for BMI and physact). (ii) Give the 95% confidence interval for this effect.

Question 1.5: (i) Based on the model in Q1.2, estimate the effect on mean HDL of a 5-unit increase in BMI (adjusting for age and physact). (ii) Give the 95% confidence interval for this effect.

```
hers <- read.dta13("/Users/anyadecarlo/biostats_208/data/final-hersdata.dta")
# Question 1.1 (i): Explore the data type of physact
# Check the structure of physact
str(hers$physact)
```

```
##  Factor w/ 5 levels "much less active",..: 4 4 3 4 5 4 4 4 1 2 ...
```

```r
# Check levels of physact
levels(hers$physact)
```

```
## [1] "much less active"     "somewhat less active" "about as active"
## [4] "somewhat more active" "much more active"
```

```r
# Frequency table of physact
table(hers$physact)
```

```
##
##     much less active somewhat less active      about as active
##                   17                   56                   77
## somewhat more active     much more active
##                   91                   35
```

```r
# Visualize the distribution of HDL by physact level
ggplot(hers, aes(x = physact, y = HDL)) +
  geom_boxplot(fill = "skyblue", color = "darkblue") +
  labs(title = "Boxplot of HDL by Physical Activity Level",
       x = "Physical Activity", y = "HDL (mg/dL)")
```

```r
# Question 1.1 (iii): Fit regression model with HDL as outcome and physact as predictor
hersLM <- lm(HDL ~ physact, data = hers)
jtools::summ(hersLM, confint = TRUE, digits = 3)
```

Observations	276
Dependent variable	HDL
Type	OLS linear regression

F(4,271)	5.767
R^2	0.078
Adj. R^2	0.065

	Est.	2.5%	97.5%	t val.	p
(Intercept)	44.941	37.994	51.889	12.735	0.000
physactsomewhat less active	-0.066	-7.999	7.866	-0.016	0.987
physactabout as active	7.111	-0.566	14.787	1.824	0.069
physactsomewhat more active	10.180	2.611	17.748	2.648	0.009
physactmuch more active	10.259	1.790	18.727	2.385	0.018

Standard errors: OLS

```r
# Calculate group means to verify model results
group_stats <- hers %>% group_by(physact) %>%
  summarise(mean_HDL = mean(HDL, na.rm = TRUE),
            sd_HDL = sd(HDL, na.rm = TRUE),
            count = n())
print(group_stats)
```

```
## # A tibble: 5 × 4
##   physact              mean_HDL sd_HDL count
##   <fct>                   <dbl>  <dbl> <int>
## 1 much less active         44.9   10.9    17
## 2 somewhat less active     44.9   10.9    56
## 3 about as active          52.1   18.4    77
## 4 somewhat more active     55.1   13.4    91
## 5 much more active         55.2   14.6    35
```

```r
# Question 1.1 (iv): Test for linear trend and departure from linear trend

# First, let's check if physact is a factor
class(hers$physact)
```

```
## [1] "factor"
```

```r
# Approach 1: Using linear regression to test for trend
levels(hers$physact)
```

```
## [1] "much less active"     "somewhat less active" "about as active"
## [4] "somewhat more active" "much more active"
```

```r
# Create a new variable with the actual numeric scores
hers$physact_score <- as.numeric(hers$physact)
# Test for trend by treating physact as a continuous predictor with actual levels
fit_trend <- lm(HDL ~ physact_score, data = hers)
jtools::summ(fit_trend, confint = TRUE, digits = 3)
```

Observations	276
Dependent variable	HDL

Type	OLS linear regression

F(1,274)	19.409
R²	0.066
Adj. R²	0.063

	Est.	2.5%	97.5%	t val.	p
(Intercept)	40.178	34.803	45.553	14.716	0.000
physact_score	3.497	1.934	5.060	4.406	0.000

Standard errors: OLS

```
# Compare with categorical model to test for departure from linearity
# Ensure physact is treated as a factor for this comparison
fit_cat <- lm(HDL ~ factor(physact), data = hers)
anova(fit_trend, fit_cat)
```

```
## Analysis of Variance Table
##
## Model 1: HDL ~ physact_score
## Model 2: HDL ~ factor(physact)
##   Res.Df   RSS Df Sum of Sq     F Pr(>F)
## 1    274 58137
## 2    271 57372  3    765.28 1.205 0.3083
```

```
# Approach 2: Using orthogonal polynomials for linear trend test
# Create orthogonal polynomial contrasts
hers$physact.cont <- hers$physact
contrasts(hers$physact.cont) <- contr.poly(nlevels(hers$physact))
```

```
# Fit model with orthogonal polynomials
hersLM_trend <- lm(HDL ~ physact.cont, data = hers)
jtools::summ(hersLM_trend, confint = TRUE, digits = 3)
```

Observations	276
Dependent variable	HDL
Type	OLS linear regression

F(4,271)	5.767
R^2	0.078
Adj. R^2	0.065

	Est.	2.5%	97.5%	t val.	p
(Intercept)	50.438	48.378	52.497	48.215	0.000
physact.cont.L	9.728	4.156	15.301	3.437	0.001
physact.cont.Q	-1.020	-6.043	4.002	-0.400	0.690
physact.cont.C	-3.236	-7.315	0.843	-1.562	0.120
physact.cont^4	1.490	-1.962	4.942	0.850	0.396

Standard errors: OLS

```
# The linear trend is represented by the first orthogonal polynomial term
# Question 1.2 (i): Adjust for age and BMI
```

```
# Fit adjusted model
hersLM_Adjusted <- lm(HDL ~ physact + age + BMI, data = hers)
jtools::summ(hersLM_Adjusted, confint = TRUE, digits = 3)
```

Observations	276
Dependent variable	HDL
Type	OLS linear regression

F(6,269)	10.448
R^2	0.189
Adj. R^2	0.171

	Est.	2.5%	97.5%	t val.	p
(Intercept)	42.739	20.096	65.382	3.716	0.000
physactsomewhat less active	1.761	-5.736	9.258	0.462	0.644
physactabout as active	6.931	-0.326	14.188	1.880	0.061
physactsomewhat more active	8.397	1.246	15.547	2.312	0.022
physactmuch more active	7.420	-0.609	15.448	1.820	0.070
age	0.359	0.101	0.618	2.737	0.007

	Est.	2.5%	97.5%	t val.	p
BMI	-0.753	-1.092	-0.415	-4.380	0.000

Standard errors: OLS

```
# Compare unadjusted and adjusted models to assess confounding
# Extract coefficients for comparison
unadj_coefs <- coef(hersLM)[grep("physact", names(coef(hersLM)))]
adj_coefs <- coef(hersLM_Adjusted)[grep("physact",
names(coef(hersLM_Adjusted)))]

# Calculate percent change in coefficients
percent_change <- (adj_coefs - unadj_coefs) / abs(unadj_coefs) * 100
print(data.frame(
  Unadjusted = unadj_coefs,
  Adjusted = adj_coefs,
  Percent_Change = percent_change

))
##                               Unadjusted Adjusted Percent_Change
## physactsomewhat less active -0.06617647 1.760995    2761.058849
## physactabout as active       7.11077158 6.931011      -2.527998
## physactsomewhat more active 10.17970265 8.396579     -17.516464
## physactmuch more active     10.25882353 7.419736     -27.674596
# Encode physact as orthogonal polynomial contrasts
hers$physact.cont <- hers$physact
contrasts(hers$physact.cont) <- contr.poly(nlevels(hers$physact))

# Fit the adjusted model with physact as a continuous term
hersLM_adjusted_trend <- lm(HDL ~ physact.cont + age + BMI, data = hers)

# Summarize the model
jtools::summ(hersLM_adjusted_trend, confint = TRUE, digits = 3)
```

Observations	276
Dependent variable	HDL
Type	OLS linear regression

F(6,269)	10.448
R^2	0.189
Adj. R^2	0.171

	Est.	2.5%	97.5%	t val.	p
(Intercept)	47.641	25.659	69.622	4.267	0.000
physact.cont.L	6.791	1.458	12.124	2.507	0.013
physact.cont.Q	-2.453	-7.218	2.311	-1.014	0.312
physact.cont.C	-1.850	-5.722	2.021	-0.941	0.348
physact.cont^4	1.001	-2.271	4.274	0.602	0.547
age	0.359	0.101	0.618	2.737	0.007
BMI	-0.753	-1.092	-0.415	-4.380	0.000

Est. 2.5% 97.5% t val. p

Standard errors: OLS

```
# Question 1.3: Estimate mean difference in HDL between "much more" active and
"about as" active


# Method 1: Using predictions with mean values of covariates
predictions(hersLM_Adjusted, by = "physact", newdata = "mean")
##
##               physact Estimate Std. Error    z Pr(>|z|)      S 2.5 % 97.5 %
##  much less active         45.7       3.33 13.7   <0.001  139.9  39.2   52.2
##  somewhat less active     47.4       1.88 25.2   <0.001  464.5  43.8   51.1
##  about as active          52.6       1.57 33.5   <0.001  813.6  49.5   55.7
##  somewhat more active     54.1       1.45 37.2   <0.001 1003.3  51.2   56.9
##  much more active         53.1       2.34 22.7   <0.001  375.3  48.5   57.7
##
## Type:  response
# Method 2: Using counterfactual approach (averaging over other predictors)
predictions(hersLM_Adjusted, by = "physact",
            newdata = datagrid(physact = levels(hers$physact),
                               grid_type = "counterfactual"))
##
##               physact Estimate Std. Error    z Pr(>|z|)      S 2.5 % 97.5 %
##  much less active         45.6       3.33 13.7   <0.001  139.4  39.0   52.1
##  somewhat less active     47.3       1.88 25.2   <0.001  464.6  43.7   51.0
##  about as active          52.5       1.58 33.3   <0.001  806.7  49.4   55.6
##  somewhat more active     54.0       1.45 37.1   <0.001 1000.4  51.1   56.8
##  much more active         53.0       2.35 22.6   <0.001  373.1  48.4   57.6
##
## Type:  response
# Check contrasts and coefficient names
contrasts(hers$physact)
##                       somewhat less active about as active somewhat more
active
```

```
## much less active                          0               0
0
## somewhat less active                      1               0
0
## about as active                           0               1
0
## somewhat more active                      0               0
1
## much more active                          0               0
0
##                       much more active
## much less active                     0
## somewhat less active                 0
## about as active                      0
## somewhat more active                 0
## much more active                     1
```

```
names(coef(hersLM_Adjusted))
```

```
## [1] "(Intercept)"                  "physactsomewhat less active"
## [3] "physactabout as active"       "physactsomewhat more active"
## [5] "physactmuch more active"      "age"
## [7] "BMI"
```

```
# Calculate the specific contrast of interest with confidence interval
hypotheses(hersLM_Adjusted, "`physactmuch more active` - `physactabout as
active`= 0", conf_level = 0.95)
```

```
##
##                                        Hypothesis Estimate Std. Error     z
##  `physactmuchmoreactive`-`physactaboutasactive`=0    0.489       2.84 0.172
##  Pr(>|z|)   S 2.5 % 97.5 %
##     0.863 0.2 -5.07   6.05
```

```
# Alternative approach using specific values
pred_grid <- datagrid(
  physact = c("much more active", "about as active"),
  age = mean(hers$age),
  BMI = mean(hers$BMI),
  model = hersLM_Adjusted
)
```

```r
# Get predictions
preds <- predictions(hersLM_Adjusted, newdata = pred_grid)


# Calculate the difference manually
much_more <- preds$estimate[preds$physact == "much more active"]
about_as <- preds$estimate[preds$physact == "about as active"]
diff <- much_more - about_as
cat("Mean difference in HDL between 'much more active' and 'about as active':",
round(diff, 3), "mg/dL\n")
```

```
## Mean difference in HDL between 'much more active' and 'about as active':
0.489 mg/dL
```

```r
# Question 1.4: Effect of 10-year decrease in age on HDL


# Calculate the effect using hypotheses function
age_effect <- hypotheses(hersLM_Adjusted, "`age` * -10 = 0", conf_level = 0.95)
print(age_effect)
```

```
##
##   Hypothesis Estimate Std. Error     z Pr(>|z|)   S 2.5 % 97.5 %
##  `age`*-10=0    -3.59       1.31 -2.74   0.0062 7.3 -6.16  -1.02
```

```r
# Alternative calculation for verification
age_coef <- coef(hersLM_Adjusted)["age"]
age_effect_manual <- age_coef * -10
age_ci <- confint(hersLM_Adjusted)["age",] * -10


cat("Effect of 10-year decrease in age on HDL:", round(age_effect_manual, 3),
    "mg/dL (95% CI:", round(age_ci[1], 3), "to", round(age_ci[2], 3), ")\n")
```

```
## Effect of 10-year decrease in age on HDL: -3.592 mg/dL (95% CI: -1.008 to
-6.176 )
```

```r
# Question 1.5: Effect of 5-unit increase in BMI on HDL


# Calculate the effect using hypotheses function
bmi_effect <- hypotheses(hersLM_Adjusted, "`BMI` * 5 = 0", conf_level = 0.95)
print(bmi_effect)
```

```
##
##  Hypothesis Estimate Std. Error     z Pr(>|z|)    S 2.5 % 97.5 %
##   `BMI`*5=0    -3.77       0.86 -4.38   <0.001 16.4 -5.45  -2.08
```

```
# Alternative calculation for verification
bmi_coef <- coef(hersLM_Adjusted)["BMI"]
bmi_effect_manual <- bmi_coef * 5
bmi_ci <- confint(hersLM_Adjusted)["BMI",] * 5

cat("Effect of 5-unit increase in BMI on HDL:", round(bmi_effect_manual, 3),
    "mg/dL (95% CI:", round(bmi_ci[1], 3), "to", round(bmi_ci[2], 3), ")\n")
```

```
## Effect of 5-unit increase in BMI on HDL: -3.765 mg/dL (95% CI: -5.458 to
-2.073 )
```

Question 2

QUESTION #2

The dataset bweight.dta contains information on 40 fictional births. The variables include birth weight in grams (birthwt), maternal age (momage) and gestational age of the infant at birth (gestationalage) in weeks. (Note that since this dataset is simulated, some of the values may fall outside realistic ranges.)

Question 2.1: Fit a regression for birth weight as a function of maternal and gestational age. Write a short description of the results

Question 2.2: Check the assumption of normally distributed residuals in this dataset; does it appear reasonable?

Question 2.3: Provide an alternate analysis that addresses violations of assumption of normality assessed in the previous question. Have your conclusions changed?

Question 2.4: Assess if there are any influential points and discuss their impact on the analysis results.

```
bweight <- read.dta13("/Users/anyadecarlo/biostats_208/data/bweight.dta")
# Question 2.1: Fit a regression for birth weight as a function of maternal
and gestational age

# Load the dataset
bweight <- read.dta13("/Users/anyadecarlo/biostats_208/data/bweight.dta")

# Examine the data structure
str(bweight)
```

```
## 'data.frame':    40 obs. of  3 variables:
##  $ gestationalage: num  30 34 32 37 32 33 37 29 35 29 ...
##  $ momage        : num  26 14 32 35 28 36 38 20 22 29 ...
```

```
##  $ birthwt       : num  3685 4180 3170 3610 3665 ...
##  - attr(*, "datalabel")= chr "Question 3 Dataset, Final Exam"
##  - attr(*, "time.stamp")= chr "10 Mar 2008 09:57"
##  - attr(*, "formats")= chr [1:3] "%9.0g" "%9.0g" "%9.0g"
##  - attr(*, "types")= int [1:3] 254 254 254
##  - attr(*, "val.labels")= Named chr [1:3] "" "" ""
##   ..- attr(*, "names")= chr [1:3] "" "" ""
##  - attr(*, "var.labels")= chr [1:3] "Gestational Age (weeks)" "Mother's Age
(years)" "Birth Weight (grams)"
##  - attr(*, "version")= int 113
##  - attr(*, "label.table")= list()
##  - attr(*, "expansion.fields")= list()
##  - attr(*, "byteorder")= int 2
##  - attr(*, "orig.dim")= int [1:2] 40 3
##  - attr(*, "data.label")= chr(0)
```

```
summary(bweight)
```

```
##  gestationalage      momage         birthwt
##  Min.   :28.00   Min.   :14.00   Min.   :1685
##  1st Qu.:30.75   1st Qu.:22.75   1st Qu.:3499
##  Median :33.00   Median :28.00   Median :3655
##  Mean   :33.55   Mean   :28.52   Mean   :3478
##  3rd Qu.:36.25   3rd Qu.:34.25   3rd Qu.:3752
##  Max.   :40.00   Max.   :39.00   Max.   :4180
```

```
# Fit regression model for birth weight
bweightLM <- lm(birthwt ~ gestationalage + momage, data = bweight)
jtools::summ(bweightLM, confint = TRUE, digits = 3)
```

Observations	40
Dependent variable	birthwt
Type	OLS linear regression

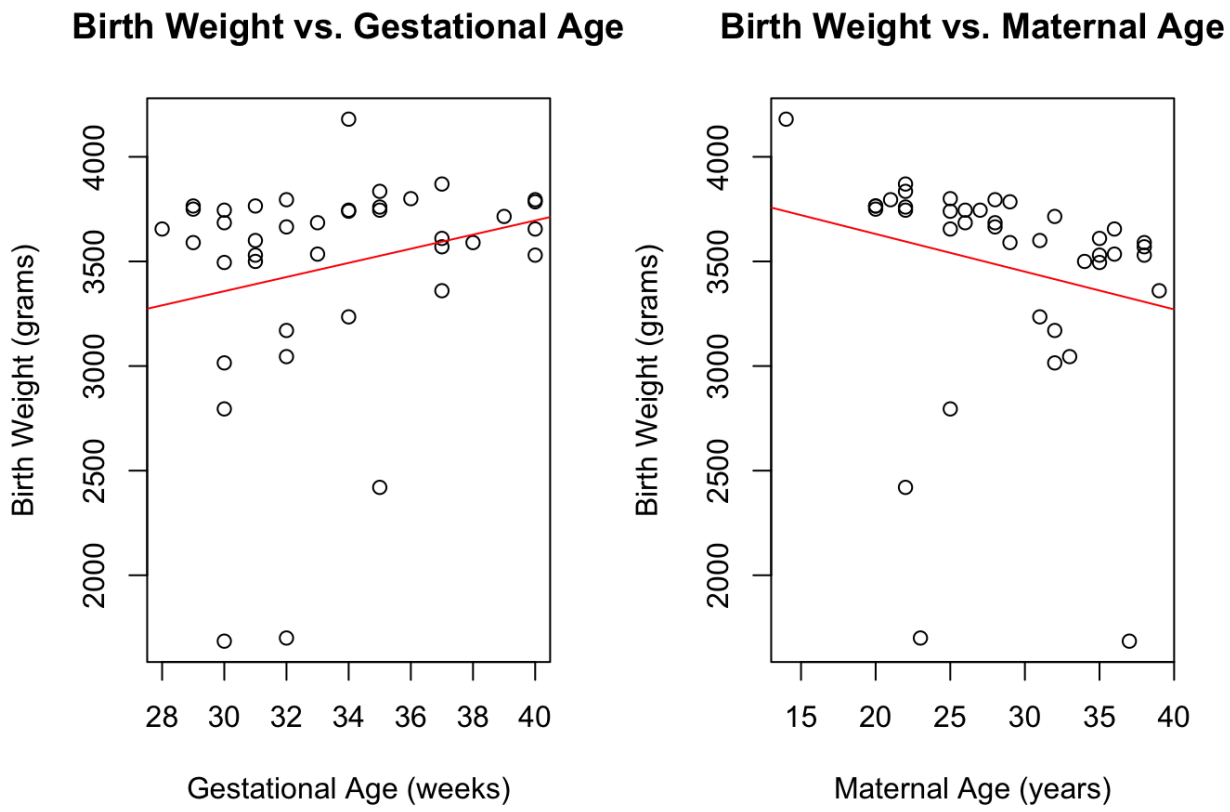
F(2,37)	3.063
R^2	0.142
Adj. R^2	0.096

	Est.	2.5%	97.5%	t val.	p
(Intercept)	2623.082	1057.826	4188.337	3.396	0.002
gestationalage	47.294	-0.389	94.977	2.010	0.052
momage	-25.664	-52.090	0.763	-1.968	0.057

Standard errors: OLS

```
# Create scatterplots to visualize relationships
par(mfrow = c(1, 2))
plot(bweight$gestationalage, bweight$birthwt,
     xlab = "Gestational Age (weeks)", ylab = "Birth Weight (grams)",
     main = "Birth Weight vs. Gestational Age")
abline(lm(birthwt ~ gestationalage, data = bweight), col = "red")

plot(bweight$momage, bweight$birthwt,
     xlab = "Maternal Age (years)", ylab = "Birth Weight (grams)",
     main = "Birth Weight vs. Maternal Age")
abline(lm(birthwt ~ momage, data = bweight), col = "red")
```




```r
par(mfrow = c(1, 1))
# Question 2.2: Check the assumption of normally distributed residuals

# Generate model diagnostics
bweight$fitted <- fitted(bweightLM)
bweight$residuals <- resid(bweightLM)

# Create comprehensive residual diagnostic plots
residual_plot <- function(data) {
  g1 <- ggplot(data, aes(x = residuals)) +
    geom_histogram(color = "black", fill = "lightblue", bins = 10) +
    labs(title = "Histogram of Residuals",
         x = "Residuals",
         y = "Count") +
    theme_minimal()
```

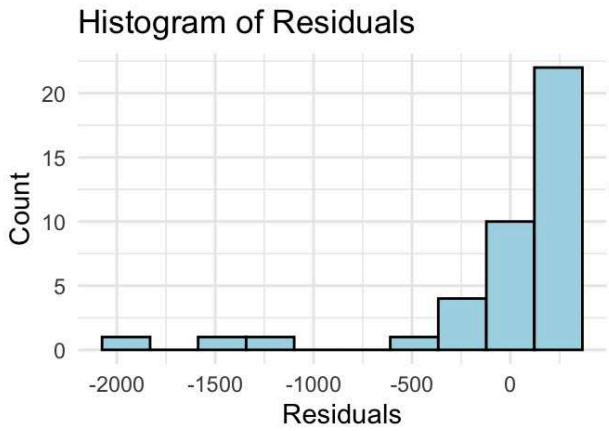
```
  g2 <- ggplot(data, aes(y = residuals)) +
    geom_boxplot(color = "black", fill = "lightblue") +
    labs(title = "Boxplot of Residuals",
         x = "",
         y = "Residuals") +
    theme_minimal()

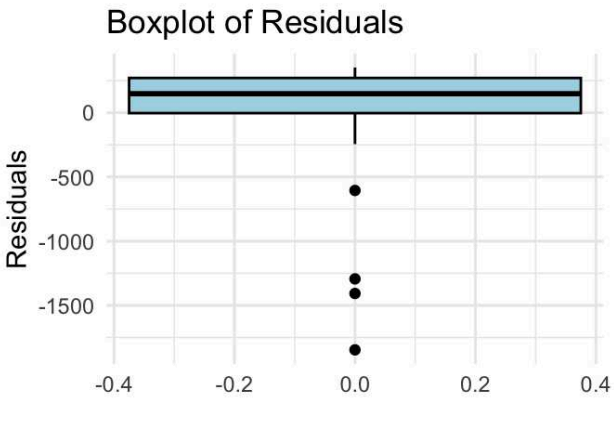
  g3 <- ggplot(data, aes(x = residuals)) +
    geom_density(color = "red") +
    stat_function(fun = dnorm,
                  args = list(mean = mean(data$residuals, na.rm = TRUE),
                              sd = sd(data$residuals, na.rm = TRUE)),
                  color = "blue") +
    labs(title = "Density Plot of Residuals",
         x = "Residuals",
         y = "Density") +
    theme_minimal()

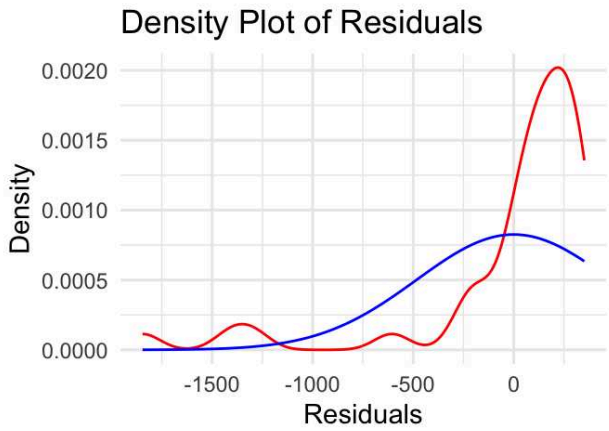
  g4 <- ggplot(data, aes(sample = residuals)) +
    stat_qq() +
    stat_qq_line() +
    labs(title = "QQ Plot of Residuals") +
    theme_minimal()

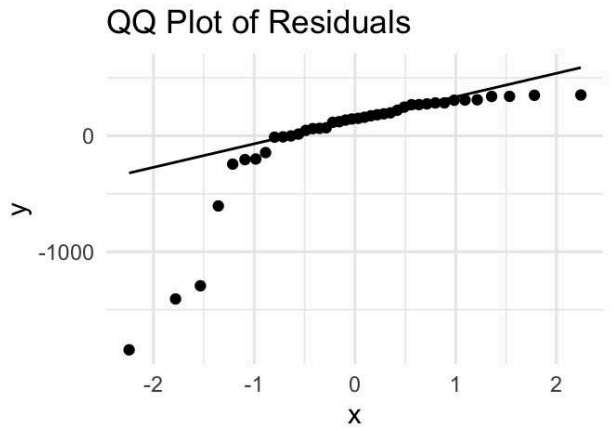
  return(grid.arrange(g1, g2, g3, g4, nrow = 2))
}

# Apply the function to the bweight dataset
residual_plot(bweight)
```










```
# Formal test of normality
shapiro_result <- shapiro.test(bweight$residuals)
print(shapiro_result)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  bweight$residuals
## W = 0.6407, p-value = 1.176e-08
```

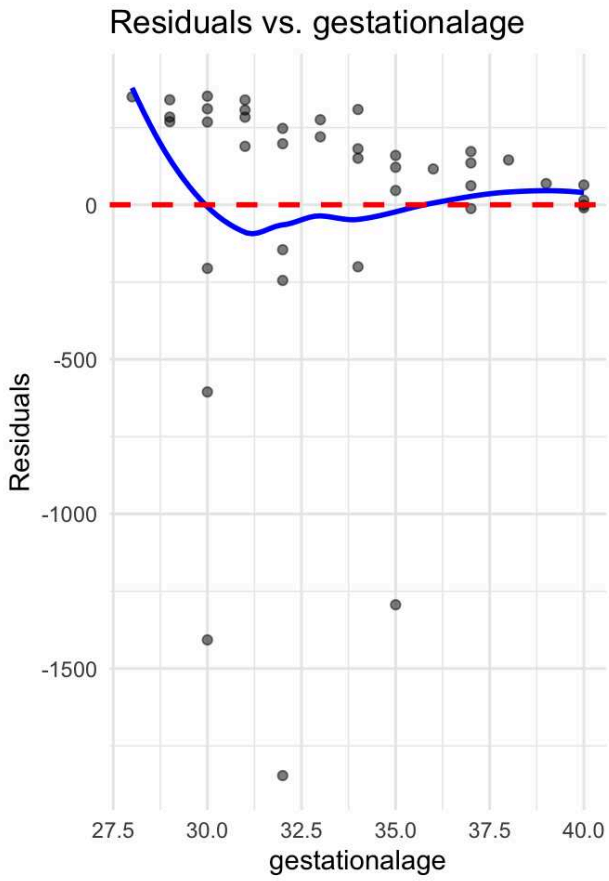
```
# Check residuals vs. fitted values (homoscedasticity)
ggplot(bweight, aes(x = fitted, y = residuals)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  geom_smooth(method = "loess", se = FALSE, color = "blue") +
  labs(title = "Residuals vs. Fitted Values",
       x = "Fitted Values",
       y = "Residuals") +
```

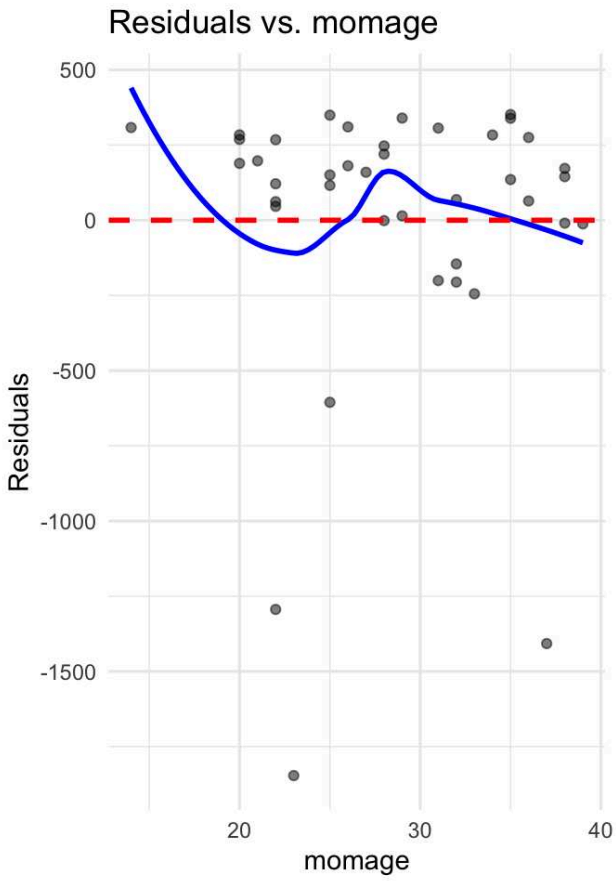
```
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```r
# Check residuals vs. predictors (linearity)
rvpPlot <- function(data, predictor) {
  ggplot(data = data, aes_string(x = predictor, y = "residuals")) +
    geom_point(alpha = 0.5) +
    geom_smooth(method = "loess", se = FALSE, color = "blue", size = 1) +
    geom_hline(yintercept = 0, linetype = "dashed", color = "red", size = 1) +
    theme_minimal() +
    labs(title = paste("Residuals vs.", predictor),
         x = predictor,
         y = "Residuals")
}

# Check residuals vs. each predictor
p1 <- rvpPlot(bweight, "gestationalage")
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## ℹ Please use tidy evaluation idioms with `aes()`.
## ℹ See also `vignette("ggplot2-in-packages")` for more information.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
p2 <- rvpPlot(bweight, "momage")
grid.arrange(p1, p2, nrow = 1)
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```






```
## Fit model with log-transformed outcome
log_model <- lm(log(birthwt) ~ gestationalage + momage, data = bweight)
jtools::summ(log_model, confint = TRUE, digits = 3)
```

Observations	40
Dependent variable	log(birthwt)
Type	OLS linear regression

F(2,37)	2.229
R²	0.108

Adj. R^2	0.059

	Est.	2.5%	97.5%	t val.	p
(Intercept)	7.824	7.241	8.407	27.188	0.000
gestationalage	0.016	-0.002	0.034	1.811	0.078
momage	-0.008	-0.017	0.002	-1.567	0.126

Standard errors: OLS

```
# Add log residuals to a copy of the dataset
bweight$log_residuals <- residuals(log_model)

# Create comprehensive residual diagnostic plots
residual_plot_log <- function(data) {
  g1 <- ggplot(data, aes(x = log_residuals)) +
    geom_histogram(color = "black", fill = "lightblue", bins = 10) +
    labs(title = "Histogram of Log Residuals",
         x = "Log Residuals",
         y = "Count") +
    theme_minimal()

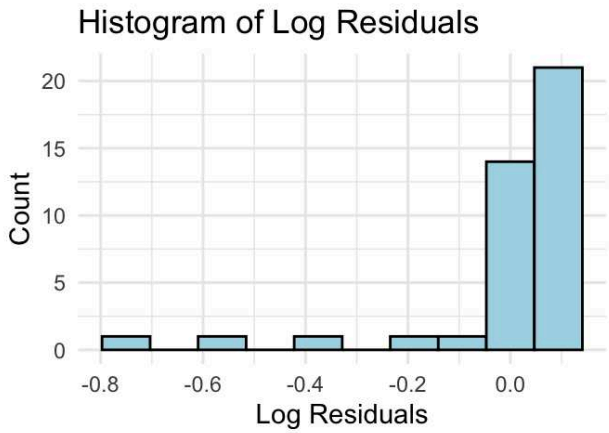
  g2 <- ggplot(data, aes(y = log_residuals)) +
    geom_boxplot(color = "black", fill = "lightblue") +
    labs(title = "Boxplot of Log Residuals",
         x = "",
         y = "Log Residuals") +
    theme_minimal()
```

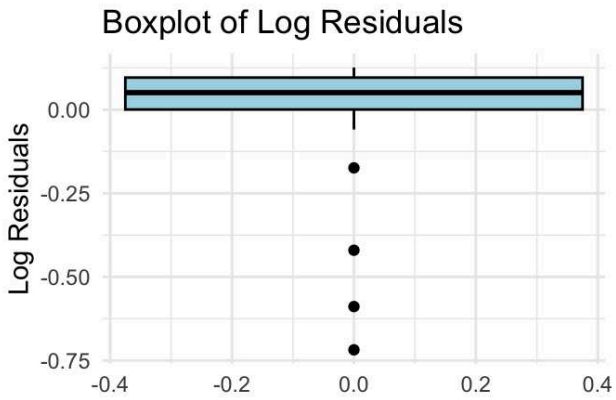
```
  g3 <- ggplot(data, aes(x = log_residuals)) +
    geom_density(color = "red") +
    stat_function(fun = dnorm,
                  args = list(mean = mean(data$log_residuals, na.rm = TRUE),
                              sd = sd(data$log_residuals, na.rm = TRUE)),
                  color = "blue") +
    labs(title = "Density Plot of Log Residuals",
         x = "Log Residuals",
         y = "Density") +
    theme_minimal()

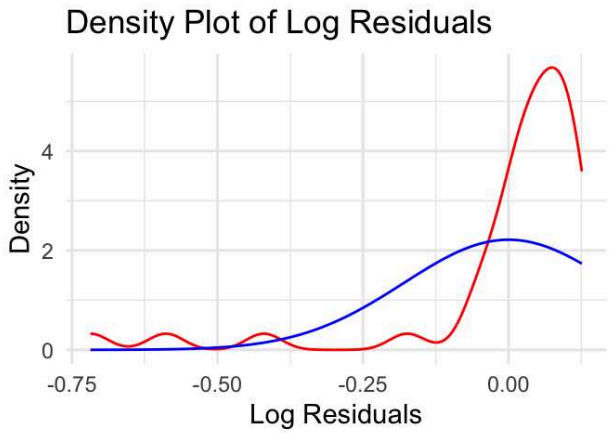
  g4 <- ggplot(data, aes(sample = log_residuals)) +
    stat_qq() +
    stat_qq_line() +
    labs(title = "QQ Plot of Log Residuals") +
    theme_minimal()

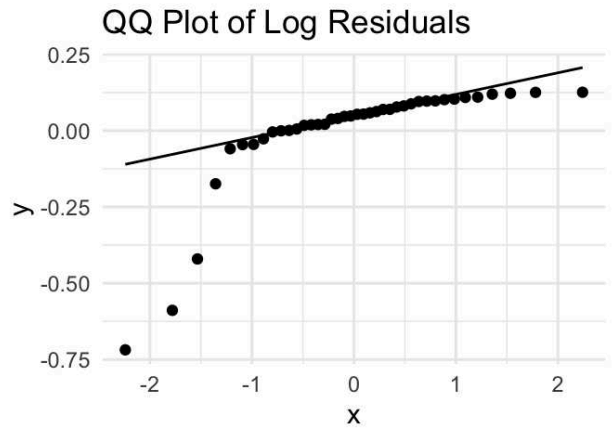
  grid.arrange(g1, g2, g3, g4, nrow = 2)
}

# Run the plotting function
residual_plot_log(bweight)
```










```
# Method 2: Robust regression with bootstrapped standard errors
set.seed(123) # For reproducibility
boot_model <- Boot(log_model, R = 1000)
boot_ci <- confint(boot_model, type = "bca")
print(boot_ci)
```

```
## Bootstrap bca confidence intervals
##
##                        2.5 %       97.5 %
## (Intercept)      7.186026000 8.106736158
## gestationalage   0.004648985 0.039711192
## momage          -0.020758877 0.001336608
```

```
# Method 3: Robust standard errors
robust_model <- lm_robust(birthwt ~ gestationalage + momage, data = bweight,
se_type = "HC3")
summary(robust_model)
```

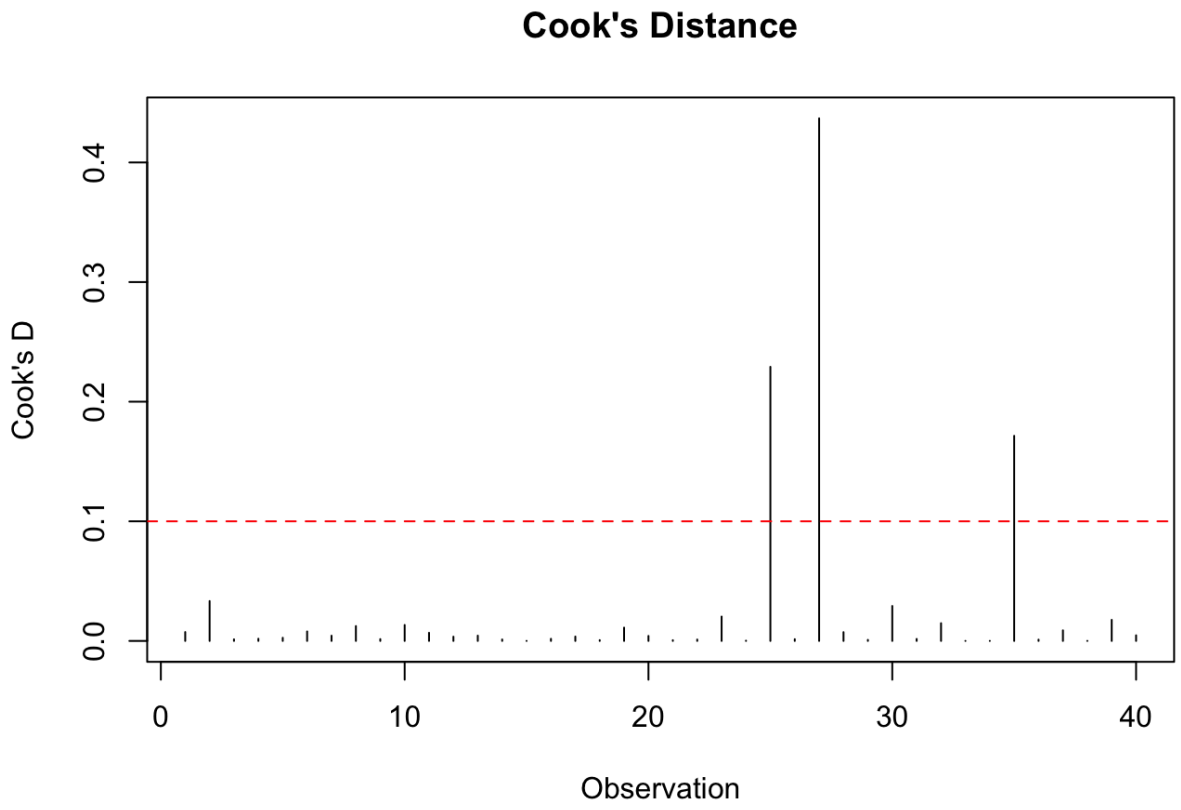
```
##
```

```
## Call:
## lm_robust(formula = birthwt ~ gestationalage + momage, data = bweight,
##     se_type = "HC3")
##
## Standard error type:  HC3
##
## Coefficients:
##                Estimate Std. Error t value  Pr(>|t|) CI Lower CI Upper DF
## (Intercept)     2623.08     591.48   4.435 7.958e-05 1424.628 3821.536 37
## gestationalage    47.29      22.59   2.094 4.317e-02    1.532   93.056 37
## momage           -25.66      16.16  -1.588 1.208e-01  -58.410    7.083 37
##
## Multiple R-squared:  0.1421 ,    Adjusted R-squared:  0.09568
## F-statistic: 2.197 on 2 and 37 DF,  p-value: 0.1254
# Compare models
cat("\nComparison of coefficient estimates:\n")
##
## Comparison of coefficient estimates:
cat("Original model (gestational age):",
round(coef(bweightLM)["gestationalage"], 2), "\n")
## Original model (gestational age): 47.29
cat("Log model (gestational age, exponentiated):",
round(exp(coef(log_model)["gestationalage"]), 2), "\n")
## Log model (gestational age, exponentiated): 1.02
cat("Robust SE model (gestational age):",
round(coef(robust_model)["gestationalage"], 2), "\n")
## Robust SE model (gestational age): 47.29
cat("\nOriginal model (maternal age):", round(coef(bweightLM)["momage"], 2),
"\n")
##
## Original model (maternal age): -25.66
cat("Log model (maternal age, exponentiated):",
round(exp(coef(log_model)["momage"]), 2), "\n")
## Log model (maternal age, exponentiated): 0.99
cat("Robust SE model (maternal age):", round(coef(robust_model)["momage"], 2),
"\n")
## Robust SE model (maternal age): -25.66
```

```r
# Question 2.4: Assess influential points and their impact


# Calculate sample size
n <- nrow(bweight)


# Calculate Cook's distance
cooks_d <- cooks.distance(bweightLM)
plot(cooks_d, type = "h", main = "Cook's Distance", ylab = "Cook's D", xlab = "Observation")
abline(h = 4/n, col = "red", lty = 2)  # Rule of thumb cutoff
```

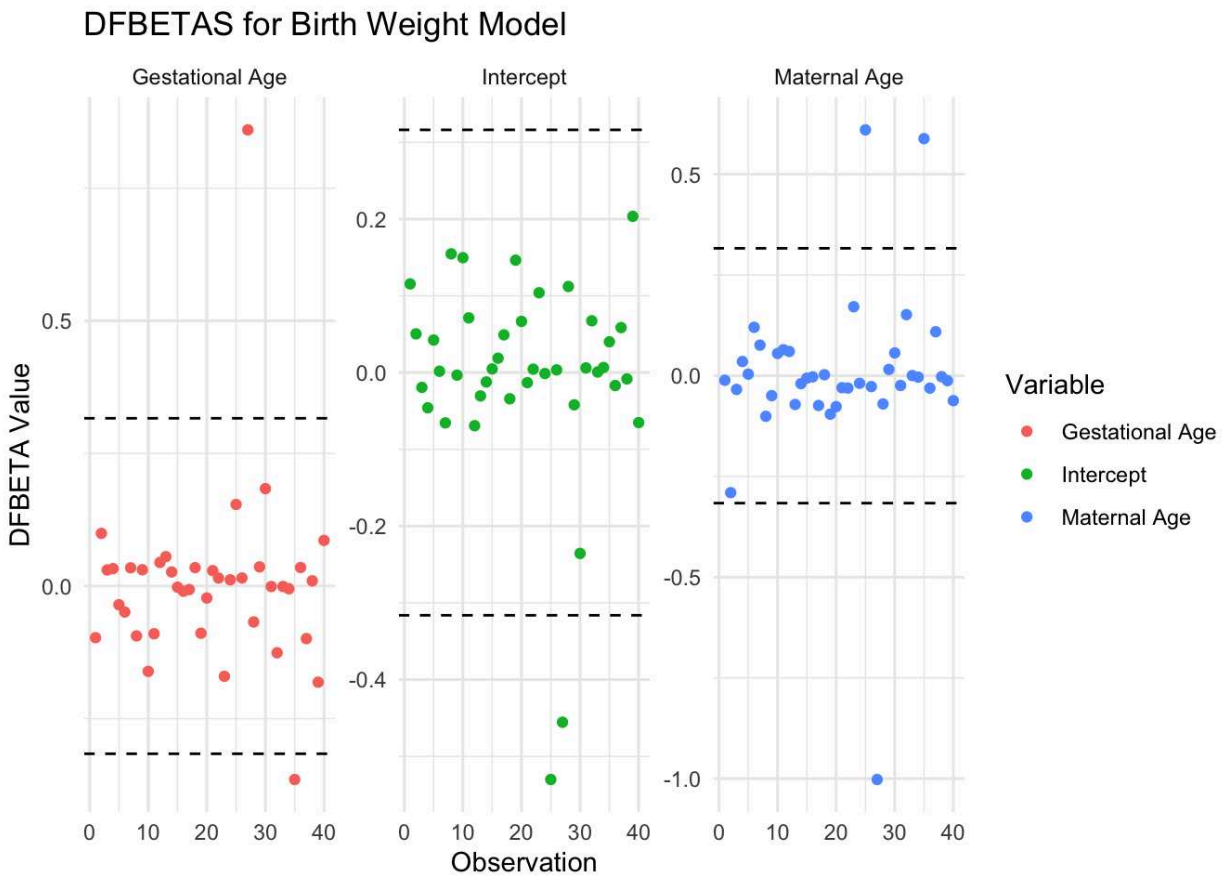



```r
# Identify influential points by Cook's distance
influential_cooks <- which(cooks_d > 4/n)
cat("Influential points by Cook's distance:", length(influential_cooks), "\n")
## Influential points by Cook's distance: 3
print(influential_cooks)
## 25 27 35
```

```
## 25 27 35
# Calculate DFBETAS
dfb <- data.frame(dfbetas(bweightLM))
ref_line <- 2/sqrt(n)  # Standard cutoff for dfbetas

# Create data frame for plotting DFBETAS
dfb_data <- data.frame(
  Variable = rep(c(
    "Intercept",
    "Gestational Age",
    "Maternal Age"
  ), each = nrow(dfb)),
  DFBETA = c(
    dfb$X.Intercept.,
    dfb$gestationalage,
    dfb$momage
  ),
  Observation = rep(1:nrow(dfb), 3)
)

# Plot DFBETAS
ggplot(dfb_data, aes(x = Observation, y = DFBETA, color = Variable)) +
  geom_point() +
  geom_hline(yintercept = c(ref_line, -ref_line), linetype = "dashed") +
  facet_wrap(~Variable, scales = "free_y") +
  labs(title = "DFBETAS for Birth Weight Model",
       x = "Observation",
       y = "DFBETA Value") +
  theme_minimal()
```




```
# Identify influential points by DFBETAS
gestage_influential <- which(abs(dfb$gestationalage) > ref_line)
momage_influential <- which(abs(dfb$momage) > ref_line)
all_influential <- unique(c(gestage_influential, momage_influential))

cat("\nInfluential points for gestational age:", length(gestage_influential),
"\n")
```

```
##
## Influential points for gestational age: 2
```

```
print(gestage_influential)
```

```
## [1] 27 35
```

```
cat("\nInfluential points for maternal age:", length(momage_influential), "\n")
```

```
##
## Influential points for maternal age: 3
```

```
print(momage_influential)
```

```
## [1] 25 27 35
```

```r
cat("\nTotal unique influential points:", length(all_influential), "\n")
```

```
##
## Total unique influential points: 3
```

```r
print(all_influential)
```

```
## [1] 27 35 25
```

```r
# Look at the values of influential points
if(length(all_influential) > 0) {
  cat("\nValues of influential observations:\n")
  print(bweight[all_influential, c("birthwt", "gestationalage", "momage")])
}
```

```
##
## Values of influential observations:
##    birthwt gestationalage momage
## 27    1685             30     37
## 35    2420             35     22
## 25    1700             32     23
```

```r
# Fit model without influential points
if(length(all_influential) > 0) {
  bweightLM_no_influential <- lm(birthwt ~ gestationalage + momage,
                                 data = bweight[-all_influential, ])

  # Compare original and robust models
  cat("\nOriginal model:\n")
  jtools::summ(bweightLM, confint = TRUE, digits = 3)

  cat("\nModel without influential points:\n")
  jtools::summ(bweightLM_no_influential, confint = TRUE, digits = 3)

  # Calculate percent changes in coefficients
  orig_coef <- coef(bweightLM)
  robust_coef <- coef(bweightLM_no_influential)
  percent_change <- (robust_coef - orig_coef) / abs(orig_coef) * 100

  cat("\nPercent change in coefficients after removing influential points:\n")
```

```
  print(percent_change)
}
##
## Original model:
##
## Model without influential points:
##
## Percent change in coefficients after removing influential points:
##    (Intercept) gestationalage         momage
##      23.081789     -25.723464      -9.964548
# Alternative approach: Use a more conservative cutoff for DFBETAS
cutoff_selected <- 0.5  # More conservative than 2/sqrt(n)
gestage_influential_alt <- which(abs(dfb$gestationalage) > cutoff_selected)
momage_influential_alt <- which(abs(dfb$momage) > cutoff_selected)
all_influential_alt <- unique(c(gestage_influential_alt,
momage_influential_alt))

if(length(all_influential_alt) > 0) {
  cat("\nUsing more conservative cutoff (", cutoff_selected, "):\n")
  cat("Total influential points:", length(all_influential_alt), "\n")

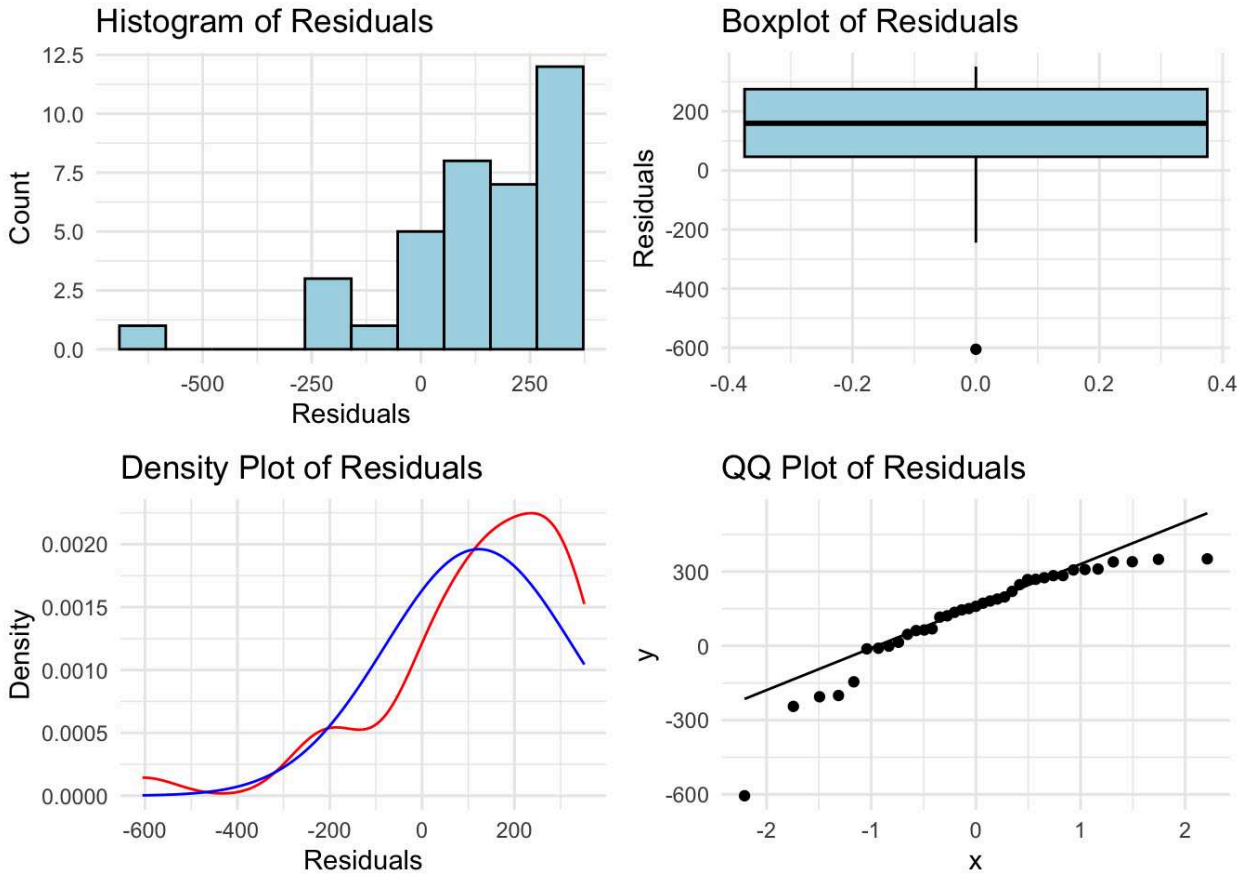
  # Fit model without these influential points
  bweightLM_no_influential_alt <- lm(birthwt ~ gestationalage + momage,
                                     data = bweight[-all_influential_alt, ])

  # Calculate percent changes in coefficients
  orig_coef <- coef(bweightLM)
  robust_coef_alt <- coef(bweightLM_no_influential_alt)
  percent_change_alt <- (robust_coef_alt - orig_coef) / abs(orig_coef) * 100

  cat("\nPercent change in coefficients after removing influential points
(conservative cutoff):\n")
  print(percent_change_alt)

}
```

```
##
## Using more conservative cutoff ( 0.5 ):
## Total influential points: 3
##
## Percent change in coefficients after removing influential points
(conservative cutoff):
##    (Intercept) gestationalage         momage
##      23.081789     -25.723464      -9.964548
# Check residuals of data set without influential points
residual_plot(bweight[-all_influential, ])
```



Question 3

QUESTION #3

This question is based on a subset of variables from the Western Collaborative Group Study (WCGS) data. The dataset final-wcgs.dta includes the following variables measured at the enrollment visit: dibpat, a binary indicator of a type-A behavior (with type-B as the reference category); smoke, a binary indicator of reporting smoking at least one cigarette per day; bmi

(body mass index); chol, total serum cholesterol in mg/dl; age (years), sbp, systolic blood pressure (mmHg); ncigs (number of cigarettes smoked per day), and npacks, a categorical indicator based on ncigs of number of packs smoked per day (with category levels corresponding to 0, < ½, ½-1, >1 – 2, >2 packs per day). The outcome variable chd69 is a binary indicator of a coronary heart disease (CHD)-related event occurring during study follow-up.

The goal is to investigate smoking, as measured by the binary indicator smoke, as a risk factor for CHD, given the available predictor variables. For questions 3.1 - 3.3, focus only on this binary measure as the predictor of primary interest - you will investigate quantitative measures of smoking (ncigs and npacks) in question 4.

Question 3.1: Write a sentence summarizing the crude (or unadjusted) effect of smoking on CHD, using the binary indicator smoke of baseline smoking status.

Question 3.2: Fit a logistic or log relative risk regression model to estimate the effect smoking on CHD adjusted for the possible confounding influence of other enrollment variables. Focus again on the smoke indicator as the predictor of primary interest, and on providing an estimate that addresses the possible causal effect of baseline smoking status on CHD risk. Assume also that any included non-categorical predictors can be adequately modeled by linear terms, and that interactions can be ignored. Answer the following questions:

i. Describe how you selected adjustment variables for your model including a discussion of what you assumed about confounding or mediating roles of the variables considered. Also, identify your chosen measure of the effect of smoking. Note that a conditional estimate (e.g. adjusted regression coefficient), or a marginal estimate (e.g. a marginal odds ratio or relative risk) is acceptable as an effect measure.
ii. Present the estimated adjusted effect measure with associated 95% confidence intervals.
iii. Provide a brief paragraph summarizing your findings, and describe issues with and/or limitations of your analysis.

Question 3.3: Fit an additional regression model including any variables adjusted for in the previous analysis, and also including a two-way interaction between the binary indicators smoke and dibpat.

i. Summarize evidence for (multiplicative) interaction from your fitted model.
ii. Present the estimated associations between the smoke variable and CHD for both type A and B behavior categories (including 95% confidence intervals).
iii. Provide a summary of evidence additive interaction between smoke and dibpat, using the relative excess risk for interaction (RERI) measure, and give the accompanying 95% confidence interval and p-value. (Note that if you used logistic regression, you may assume that the outcome is sufficiently rare that the odds ratio-based version of RERI is acceptable.)

```
# Read in data

wcgs <- read.dta13("/Users/anyadecarlo/biostats_208/data/final-wcgs.dta",
nonint.factors = TRUE)
```

```
# Table summary of variables with variable classification
summary(wcgs)
```

```
##        id              age              sbp              chol           ncigs
##  Min.   : 2001   Min.   :39.00   Min.   : 98.0   Min.   :103.0   Min.   :
0.0
##  1st Qu.: 3741   1st Qu.:42.00   1st Qu.:120.0   1st Qu.:197.2   1st Qu.:
0.0
##  Median :11406   Median :45.00   Median :126.0   Median :223.0   Median :
0.0
##  Mean   :10478   Mean   :46.28   Mean   :128.6   Mean   :226.4   Mean
:11.6
##  3rd Qu.:13115   3rd Qu.:50.00   3rd Qu.:136.0   3rd Qu.:253.0   3rd
Qu.:20.0
##  Max.   :22101   Max.   :59.00   Max.   :230.0   Max.   :645.0   Max.
:99.0
##                                                  NA's   :12
##    dibpat     chd69           bmi               smoke        npacks
##  B3,B4:1565   no :2897   Min.   :11.19   nonsmoker:1652   0    :1652
##  A1,A2:1589   yes: 257   1st Qu.:22.96   smoker   :1502   <1/2 : 105
##                          Median :24.39                    1/2-1: 700
##                          Mean   :24.52                    >1-2 : 634
##                          3rd Qu.:25.84                    >2   :  63
##                          Max.   :38.95
##
```

```
# Check classes
str(wcgs)
```

```
## 'data.frame':    3154 obs. of  10 variables:
##  $ id    : num  2001 2002 2003 2004 2005 ...
##  $ age   : num  49 42 42 41 59 44 44 40 43 42 ...
##  $ sbp   : num  110 154 110 124 144 150 130 138 146 132 ...
##  $ chol  : num  225 177 181 132 255 182 155 140 149 325 ...
##  $ ncigs : num  25 20 0 20 20 0 0 0 25 0 ...
##  $ dibpat: Factor w/ 2 levels "B3,B4","A1,A2": 2 2 1 1 1 1 1 2 1 2 ...
##  $ chd69 : Factor w/ 2 levels "no","yes": 1 1 1 1 2 1 1 1 1 1 ...
##  $ bmi   : num  19.8 23 23.6 23.1 21.5 ...
##  $ smoke : Factor w/ 2 levels "nonsmoker","smoker": 2 2 1 2 2 1 1 1 2 1 ...
##  $ npacks: Factor w/ 5 levels "0","<1/2","1/2-1",..: 4 3 1 3 3 1 1 1 4 1 ...
```

```
##  - attr(*, "datalabel")= chr ""
##  - attr(*, "time.stamp")= chr "12 Mar 2017 14:58"
##  - attr(*, "formats")= chr [1:10] "%9.0g" "%9.0g" "%9.0g" "%9.0g" ...
##  - attr(*, "types")= int [1:10] 65527 65527 65527 65527 65527 65527 65527
65527 65527 65527
##  - attr(*, "val.labels")= Named chr [1:10] "" "" "" "" ...
##   ..- attr(*, "names")= chr [1:10] "" "" "" "" ...
##  - attr(*, "var.labels")= chr [1:10] "id" "age" "systolic BP" "total
cholesterol" ...
##  - attr(*, "version")= int 118
##  - attr(*, "label.table")=List of 4
##   ..$ bpat2 : Named int [1:2] 0 1
##   .. ..- attr(*, "names")= chr [1:2] "B3,B4" "A1,A2"
##   ..$ ind   : Named int [1:2] 0 1
##   .. ..- attr(*, "names")= chr [1:2] "no" "yes"
##   ..$ smlab : Named int [1:2] 0 1
##   .. ..- attr(*, "names")= chr [1:2] "nonsmoker" "smoker"
##   ..$ smlab2: Named int [1:5] 0 1 2 3 4
##   .. ..- attr(*, "names")= chr [1:5] "0" "<1/2" "1/2-1" ">1-2" ...
##  - attr(*, "expansion.fields")= list()
##  - attr(*, "byteorder")= chr "LSF"
##  - attr(*, "orig.dim")= int [1:2] 3154 10
##  - attr(*, "data.label")= chr(0)
```

```
# Question 3.1: Crude (unadjusted) effect of smoking on CHD


# Crude Odds Ratio for effect of smoking on CHD
model <- glm(chd69 ~ smoke, family = binomial, data = wcgs)
jtools::summ(model, confint = TRUE, digits = 3, exp = TRUE)
```

Observations	3154
Dependent variable	chd69
Type	Generalized linear model

Family	binomial
Link	logit

χ^2(1)	22.863
p	0.000
Pseudo-R² (Cragg-Uhler)	0.017
Pseudo-R² (McFadden)	0.013
AIC	1762.381
BIC	1774.494

	exp(Est.)	2.5%	97.5%	z val.	p
(Intercept)	0.063	0.051	0.077	-26.535	0.000
smokesmoker	1.877	1.445	2.440	4.710	0.000

Standard errors:
MLE

```
# Unadjusted odds ratio for effect of smoking on CHD
oddsratio(wcgs$chd69, wcgs$smoke)
## $data
##          Outcome
## Predictor nonsmoker smoker Total
##     no         1554   1343  2897
##     yes          98    159   257
```

```
##     Total      1652   1502  3154
##
## $measure
##          odds ratio with 95% C.I.
## Predictor estimate    lower    upper
##       no  1.000000       NA       NA
##       yes 1.875857 1.445907 2.445018
##
## $p.value
##          two-sided
## Predictor   midp.exact fisher.exact   chi.square
##       no            NA           NA           NA
##       yes 1.834284e-06 2.365647e-06 1.831404e-06
##
## $correction
## [1] FALSE
##
## attr(,"method")
## [1] "median-unbiased estimate & mid-p exact CI"
```

```
# Alternative approach using epitools
epitab(wcgs$smoke, wcgs$chd69, method = "oddsratio")
```

```
## $tab
##            Outcome
## Predictor     no       p0 yes       p1 oddsratio    lower   upper
p.value
##   nonsmoker 1554 0.536417  98 0.381323  1.000000       NA      NA
NA
##   smoker    1343 0.463583 159 0.618677  1.877353 1.444515 2.43989
2.365647e-06
##
## $measure
## [1] "wald"
##
## $conf.level
## [1] 0.95
##
```

```
## $pvalue
## [1] "fisher.exact"
```

```
# Create cross-tabulation and visualization of smoking and CHD
# Let's use a simpler approach with a table
smoke_chd_table <- with(wcgs, table(smoke, chd69))
print(smoke_chd_table)
```

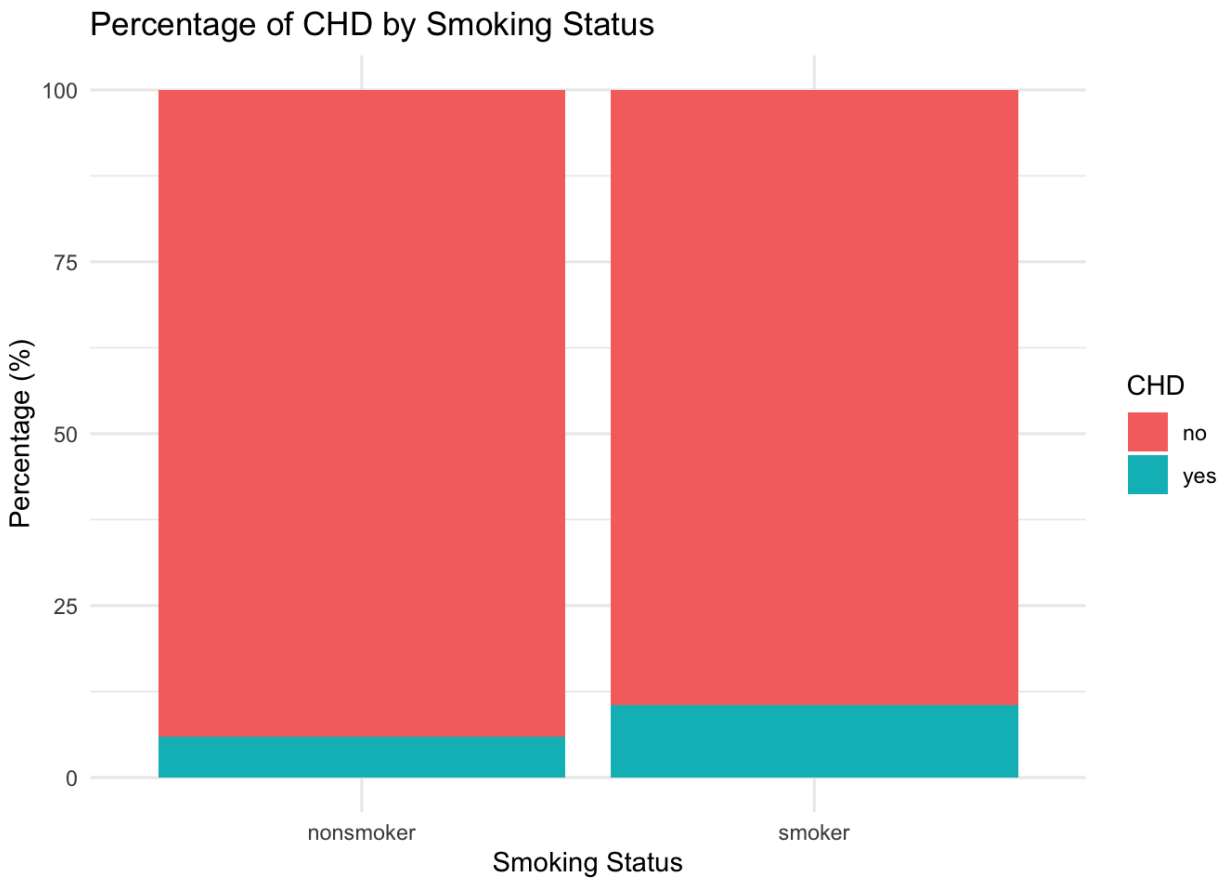
```
##            chd69
## smoke         no  yes
##   nonsmoker 1554   98
##   smoker    1343  159
```

```
# Calculate proportions
prop_table <- prop.table(smoke_chd_table, margin = 1) * 100
print(prop_table)
```

```
##            chd69
## smoke              no       yes
##   nonsmoker 94.067797  5.932203
##   smoker    89.414115 10.585885
```

```
# Create a data frame for visualization
viz_data <- as.data.frame(prop_table)
names(viz_data) <- c("Smoking", "CHD", "Percentage")

# Visualize the crude relationship
ggplot(viz_data, aes(x = Smoking,
                     y = Percentage,
                     fill = CHD)) +
  geom_bar(stat = "identity", position = "stack") +
  labs(title = "Percentage of CHD by Smoking Status",
       x = "Smoking Status",
       y = "Percentage (%)") +
  theme_minimal()
```



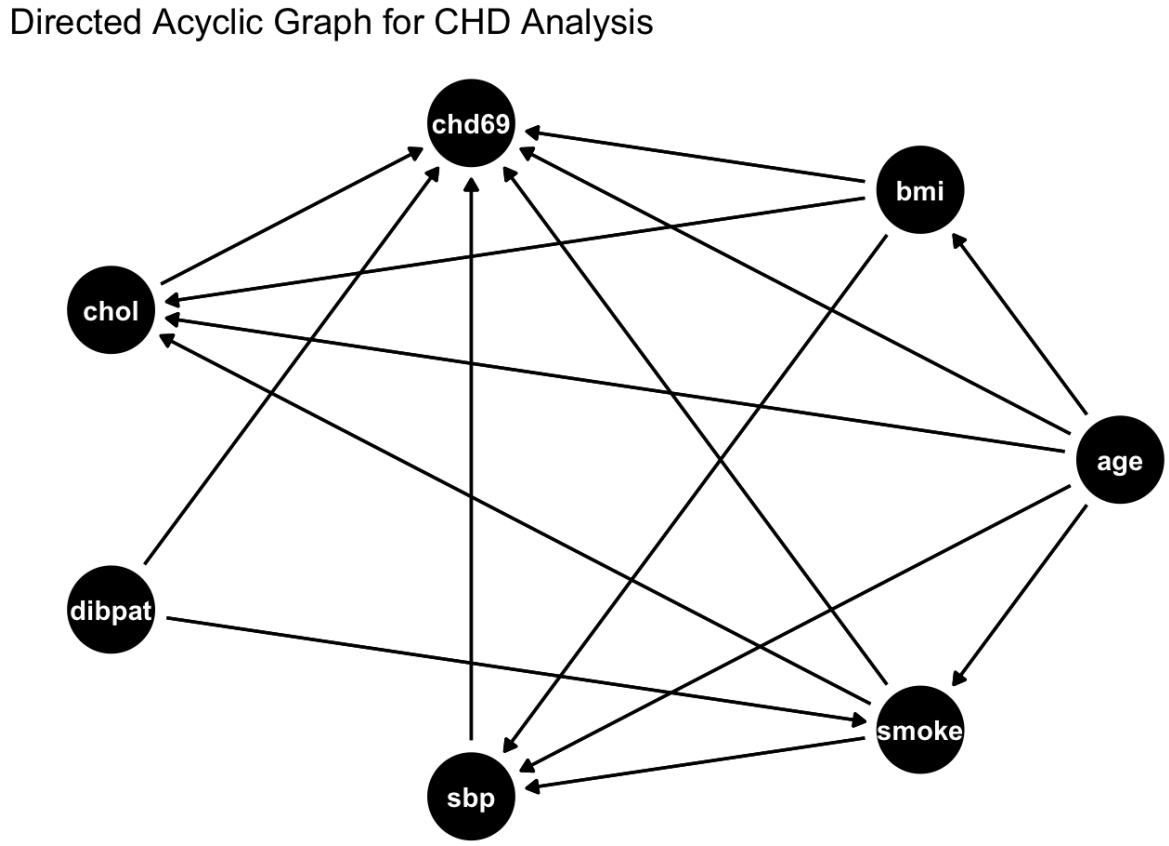

```
# Question 3.2: Adjusted effect of smoking on CHD - Part 1: DAG and confounder selection


# Create DAG based on subject matter knowledge
library(dagitty)
library(ggdag)


dag <- dagitty("dag {
  smoke -> chd69 [exposure]
  age -> chd69
  age -> smoke
  dibpat -> smoke
  dibpat -> chd69
  bmi -> chd69
  chol -> chd69
  sbp -> chd69
```

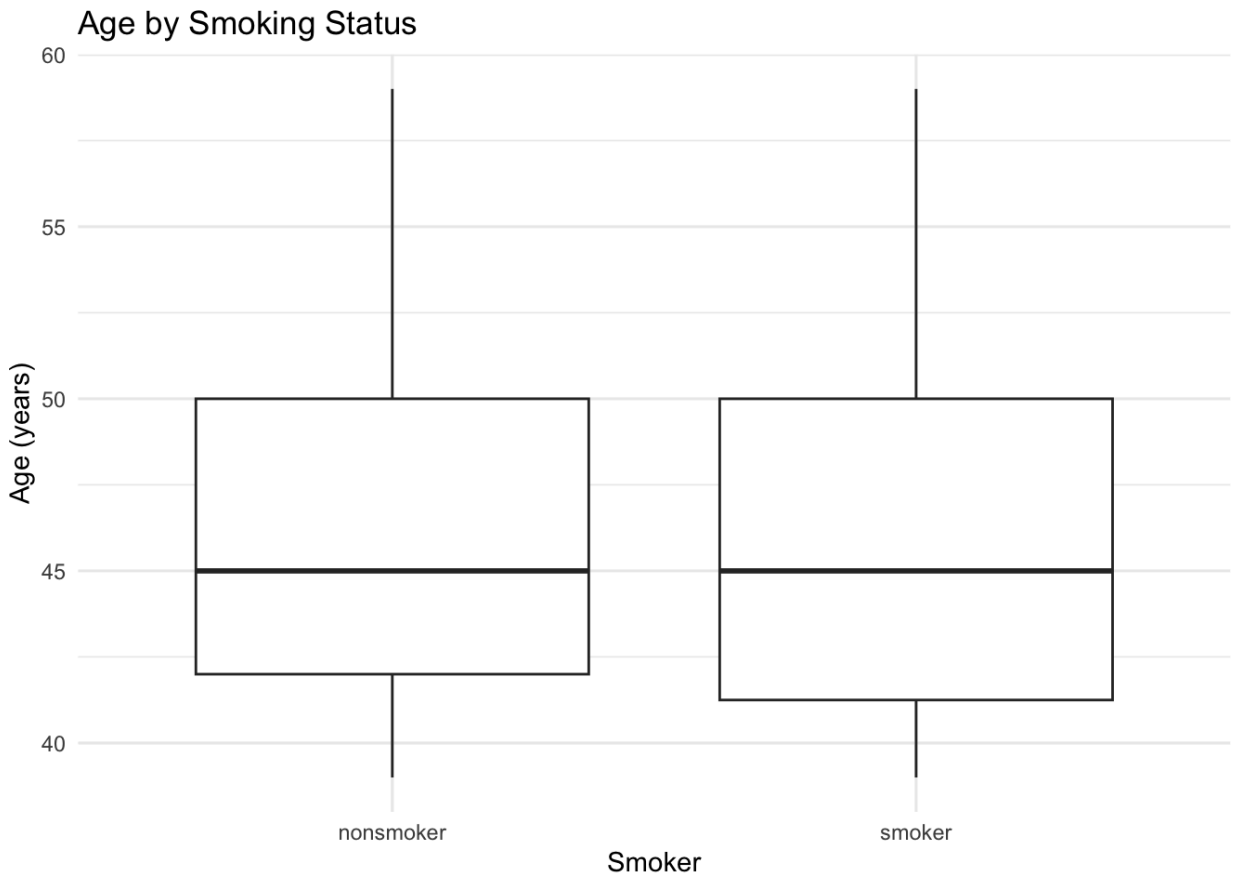
```
  age -> bmi
  age -> sbp
  age -> chol
  bmi -> sbp
  bmi -> chol
  smoke -> sbp
  smoke -> chol
}")


# Plot the DAG
ggdag(dag, layout = "circle") +
  theme_dag() +
  geom_dag_edges(arrow_directed = grid::arrow(length = grid::unit(5, "pt"),
type = "closed")) +
  ggtitle("Directed Acyclic Graph for CHD Analysis")
```

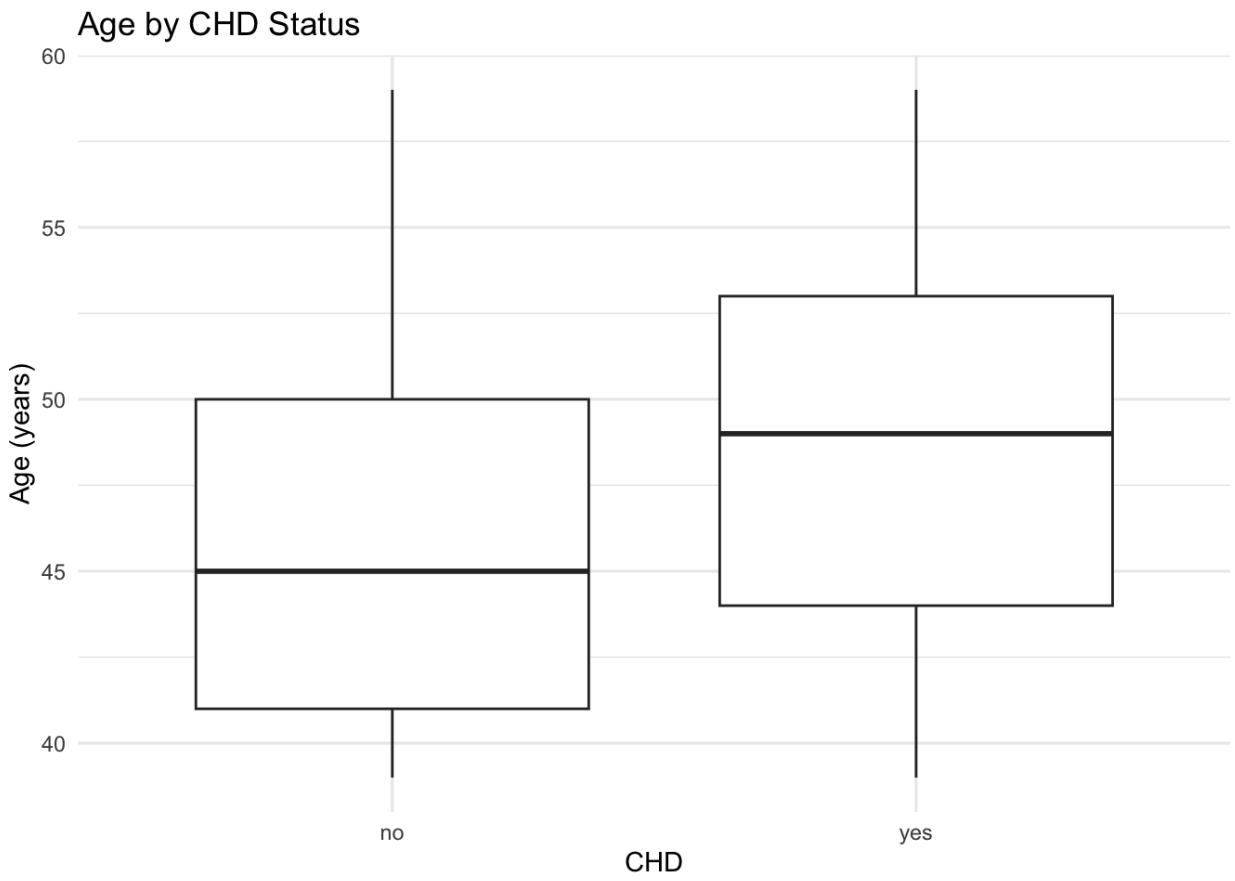



```
# Identify adjustment sets for estimating the effect of smoking on CHD
```

```
adjustmentSets(dag, exposure = "smoke", outcome = "chd69")
## { age, dibpat }
# Explore relationships between potential confounders and exposure/outcome
# Age vs. smoking
ggplot(wcgs, aes(x = smoke, y = age)) +
  geom_boxplot() +
  labs(title = "Age by Smoking Status", x = "Smoker", y = "Age (years)") +
  theme_minimal()
```




```
# Age vs. CHD
ggplot(wcgs, aes(x = chd69, y = age)) +
  geom_boxplot() +
  labs(title = "Age by CHD Status", x = "CHD", y = "Age (years)") +
  theme_minimal()
```




```
# Behavior pattern vs. smoking
table(wcgs$dibpat, wcgs$smoke)
```

```
##
##          nonsmoker smoker
##   B3,B4        868    697
##   A1,A2        784    805
```

```
prop.table(table(wcgs$dibpat, wcgs$smoke), 1) * 100
```

```
##
##          nonsmoker   smoker
##   B3,B4  55.46326 44.53674
##   A1,A2  49.33921 50.66079
```

```
# Behavior pattern vs. CHD
table(wcgs$dibpat, wcgs$chd69)
```

```
##
##            no  yes
##   B3,B4 1486   79
```

```
##   A1,A2 1411  178
```

```
prop.table(table(wcgs$dibpat, wcgs$chd69), 1) * 100
```

```
##
##                no       yes
##   B3,B4 94.952077  5.047923
##   A1,A2 88.797986 11.202014
```

```
# Question 3.2: Adjusted effect of smoking on CHD - Part 3: Model fitting and interpretation


# Unadjusted model
model_unadj <- glm(chd69 ~ smoke, family = binomial, data = wcgs)


# Adjusted model
model_adjust <- glm(chd69 ~ smoke + age + dibpat,
                    family = binomial,
                    data = wcgs)
jtools::summ(model_adjust, confint = TRUE, digits = 3, exp = TRUE)
```

Observations	3154
Dependent variable	chd69
Type	Generalized linear model
Family	binomial
Link	logit

$\chi^2(3)$	97.348
p	0.000
Pseudo-R^2 (Cragg-Uhler)	0.070

Pseudo-R² (McFadden)	0.055
AIC	1691.895
BIC	1716.121

	exp(Est.)	2.5%	97.5%	z val.	p
(Intercept)	0.001	0.000	0.005	-11.480	0.000
smokesmoker	1.825	1.399	2.381	4.440	0.000
age	1.072	1.049	1.097	6.092	0.000
dibpatA1,A2	2.158	1.635	2.850	5.426	0.000

Standard errors:
MLE

```
# Visualize adjusted predictions
newdata <- expand.grid(
  smoke = factor(c("nonsmoker", "smoker"), levels = c("nonsmoker", "smoker")),
  dibpat = factor(c("B3,B4", "A1,A2"), levels = c("B3,B4", "A1,A2")),  # Include both behavior types
  age = mean(wcgs$age)
)

# Add adjusted predicted probabilities
newdata$pred_adj <- predict(model_adjust, newdata = newdata, type = "response")

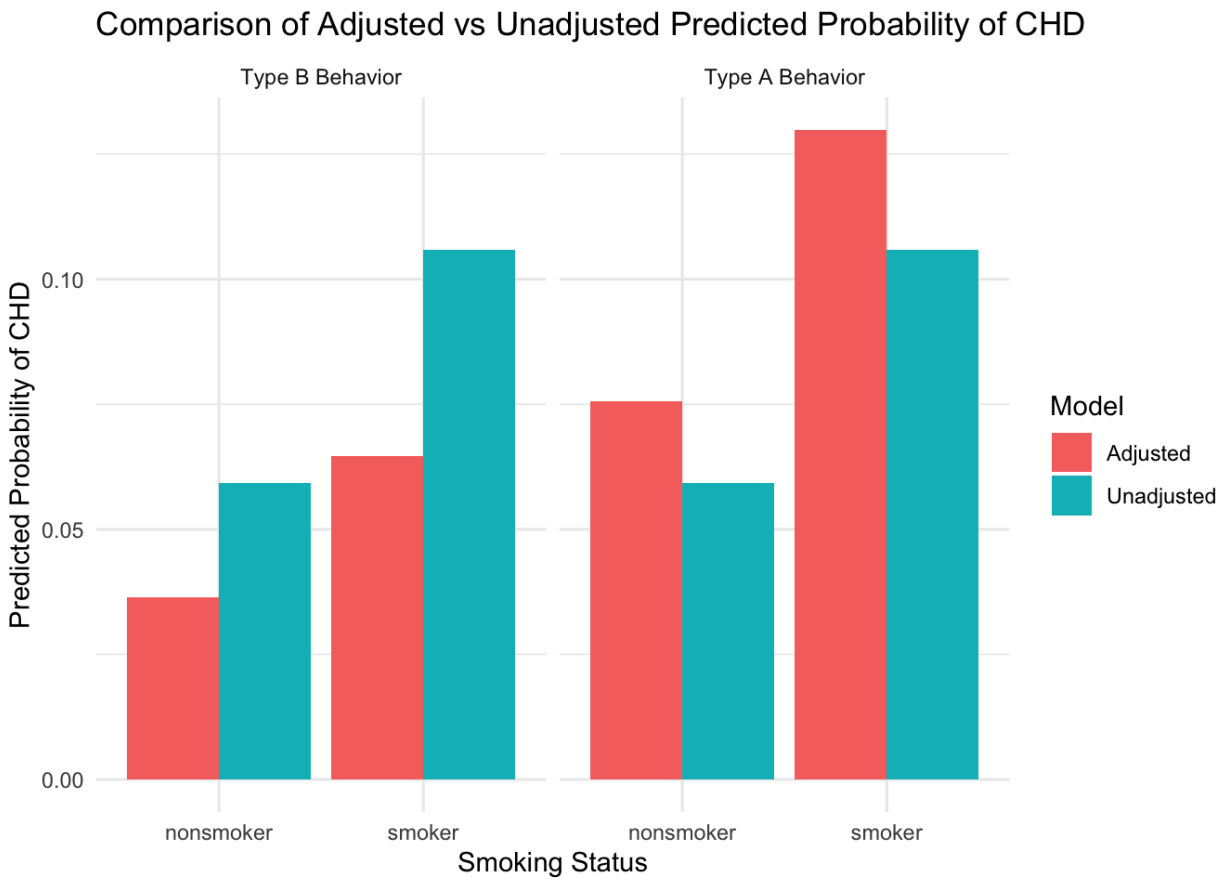
# Add unadjusted predicted probabilities
# For unadjusted model, create predictions for unique smoking values
unadj_preds <- predict(model_unadj,
                       newdata = data.frame(smoke = levels(newdata$smoke)),
```

```
                        type = "response")

# Map these predictions back based on smoking status
newdata$pred_unadj <- unadj_preds[as.numeric(newdata$smoke)]

# Convert to long format for easier plotting
library(tidyr)
newdata_long <- pivot_longer(
  newdata,
  cols = c(pred_adj, pred_unadj),
  names_to = "model",
  values_to = "probability"
)

# Plot both adjusted and unadjusted predictions
ggplot(newdata_long, aes(x = smoke, y = probability, fill = model)) +
  geom_bar(stat = "identity", position = "dodge") +
  facet_wrap(~ dibpat, labeller = labeller(dibpat = c("B3,B4" = "Type B
Behavior", "A1,A2" = "Type A Behavior"))) +
  scale_fill_discrete(name = "Model", labels = c("Adjusted", "Unadjusted")) +
  labs(title = "Comparison of Adjusted vs Unadjusted Predicted Probability of
CHD",
       x = "Smoking Status",
       y = "Predicted Probability of CHD") +
  theme_minimal()
```




```
# Question 3.2(iii): Provide a brief paragraph summarizing your findings, and describe issues with and/or limitations of your analysis.


# 1. Hypothesis test comparing unadjusted and adjusted models

# Unadjusted model (already defined earlier)

# Adjusted model (already defined earlier)

model_adjust <- glm(chd69 ~ smoke + age + dibpat, family = binomial, data = wcgs)


# Compare models using likelihood ratio test

anova(model_unadj, model_adjust, test = "Chisq")
```

```
## Analysis of Deviance Table

##

## Model 1: chd69 ~ smoke

## Model 2: chd69 ~ smoke + age + dibpat

##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
```

```
## 1      3152     1758.4
## 2      3150     1683.9  2   74.486 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Extract odds ratios from both models for comparison
exp(coef(model_unadj)["smokesmoker"])  # Unadjusted OR for smoking
```

```
## smokesmoker
##    1.877353
```

```
exp(coef(model_adjust)["smokesmoker"]) # Adjusted OR for smoking
```

```
## smokesmoker
##    1.825351
```

```
# Compare predicted probabilities from adjusted vs unadjusted models
# Create prediction data
pred_compare <- expand.grid(
  smoke = factor(c("nonsmoker", "smoker"), levels = c("nonsmoker", "smoker")),
  age = c(40, 50, 60)
)

# Add behavior pattern (for adjusted model only)
pred_compare$dibpat <- factor("B3,B4", levels = c("B3,B4", "A1,A2"))

# Get predictions from both models
pred_compare$prob_unadj <- predict(model_unadj, newdata = pred_compare, type =
"response")
pred_compare$prob_adj <- predict(model_adjust, newdata = pred_compare, type =
"response")

# Convert to long format for easier plotting
library(tidyr)
pred_long <- pivot_longer(
  pred_compare,
  cols = c(prob_unadj, prob_adj),
  names_to = "model",
  values_to = "probability"
)
```

```r
# Visualize the comparison
ggplot(pred_long, aes(x = smoke, y = probability, fill = model)) +
  geom_bar(stat = "identity", position = "dodge") +
  facet_wrap(~ age, labeller = labeller(age = function(x) paste("Age", x))) +
  labs(title = "Comparison of Adjusted vs Unadjusted Predicted Probability of CHD",
       x = "Smoking Status",
       y = "Predicted Probability of CHD") +
  theme_minimal()
```

```r
# Question 3.3: Interaction between smoking and behavior pattern

# Fit model with interaction between smoke and dibpat
model_interaction <- glm(chd69 ~ smoke * dibpat + age,
                         family = binomial,
                         data = wcgs)

# Display model summary with odds ratios
jtools::summ(model_interaction, confint = TRUE, digits = 3, exp = TRUE)
```

Observations	3154
Dependent variable	chd69
Type	Generalized linear model
Family	binomial
Link	logit

$\chi^2(4)$	99.117
p	0.000
Pseudo-R^2 (Cragg-Uhler)	0.072
Pseudo-R^2 (McFadden)	0.056
AIC	1692.126
BIC	1722.408

	exp(Est.)	2.5%	97.5%	z val.	p
(Intercept)	0.001	0.000	0.004	-11.482	0.000
smokesmoker	2.370	1.476	3.808	3.569	0.000
dibpatA1,A2	2.729	1.736	4.288	4.353	0.000
age	1.072	1.048	1.096	6.074	0.000
smokesmoker:dibpatA1,A2	0.680	0.383	1.205	-1.322	0.186

Standard errors: MLE

```
# Use marginaleffects to calculate odds ratios by behavior pattern
library(marginaleffects)

# Calculate average marginal effects of smoking within each behavior pattern
smoke_effects <- avg_comparisons(
  model_interaction,
  variables = "smoke",
  by = "dibpat",
  comparison = "lnor",  # log odds ratio
  transform = exp       # exponentiate to get odds ratios
)

# Print the results
print(smoke_effects)
##
##  dibpat Estimate Pr(>|z|)    S 2.5 % 97.5 %
##   B3,B4     2.35  < 0.001 11.4  1.47   3.77
##   A1,A2     1.60  0.00362  8.1  1.17   2.19
##
## Term: smoke
```

```
## Type:  response
## Comparison: ln(odds(smoker) / odds(nonsmoker))
```

```
# Test for multiplicative interaction
# Extract the interaction coefficient and its p-value
interaction_coef <- coef(summary(model_interaction))["smokesmoker:dibpatA1,A2"]
cat("Multiplicative interaction coefficient (log odds scale):",
    round(interaction_coef[1], 3),
    ", p-value:",
    round(interaction_coef[4], 3))
```

```
## Multiplicative interaction coefficient (log odds scale): NA , p-value: NA
```

```
# Test the null hypothesis that the interaction coefficient is zero
hypotheses(model_interaction, "`smokesmoker:dibpatA1,A2` = 0", transform = exp)
```

```
##
##                   Hypothesis Estimate Std. Error     z Pr(>|z|)   S  2.5 %
##  `smokesmoker:dibpatA1,A2`=0   -0.386      0.292 -1.32    0.186 2.4 -0.959
##  97.5 %
##   0.186
```

```
# Test the effect of smoking for Type B behavior
hypotheses(model_interaction, "`smokesmoker` = 0", transform = exp)
```

```
##
##       Hypothesis Estimate Std. Error    z Pr(>|z|)    S 2.5 % 97.5 %
##  `smokesmoker`=0    0.863      0.242 3.57   <0.001 11.4 0.389   1.34
```

```
# Effect of smoking for Type A behavior
hypotheses(model_interaction, "`smokesmoker` + `smokesmoker:dibpatA1,A2` = 0",
transform = exp)
```

```
##
##                                 Hypothesis Estimate Std. Error    z Pr(>|z|)
##  `smokesmoker`+`smokesmoker:dibpatA1,A2`=0    0.477      0.164 2.91  0.00362
##    S 2.5 % 97.5 %
##  8.1 0.156  0.798
```

```
# Calculate RERI (Relative Excess Risk due to Interaction) using hypotheses
function
reri_result <- hypotheses(model_interaction,
         "exp(`smokesmoker` + `dibpatA1,A2` + `smokesmoker:dibpatA1,A2`) -
exp(`smokesmoker`) - exp(`dibpatA1,A2`) + 1 = 0")
print(reri_result)
```

```
##
##
Hypothesis
##
exp(`smokesmoker`+`dibpatA1,A2`+`smokesmoker:dibpatA1,A2`)-exp(`smokesmoker`)-e
xp(`dibpatA1,A2`)+1=0
##  Estimate Std. Error     z Pr(>|z|)   S 2.5 % 97.5 %
##     0.297      0.684 0.434    0.665 0.6 -1.04   1.64
# Alternative calculation of RERI for verification
or_11 <- exp(coef(model_interaction)["smokesmoker"] +
             coef(model_interaction)["dibpatA1,A2"] +
             coef(model_interaction)["smokesmoker:dibpatA1,A2"])
or_10 <- exp(coef(model_interaction)["smokesmoker"])
or_01 <- exp(coef(model_interaction)["dibpatA1,A2"])
reri <- or_11 - or_10 - or_01 + 1


cat("RERI =", round(reri, 3), "\n")
## RERI = 0.297
cat("RERI > 0 suggests positive additive interaction\n")
## RERI > 0 suggests positive additive interaction
cat("RERI = 0 suggests no additive interaction\n")
## RERI = 0 suggests no additive interaction
cat("RERI < 0 suggests negative additive interaction\n")
## RERI < 0 suggests negative additive interaction
# Visualize the interaction
newdata_int <- expand.grid(
  smoke = factor(c("nonsmoker", "smoker"), levels = c("nonsmoker", "smoker")),
  dibpat = factor(c("B3,B4", "A1,A2"), levels = c("B3,B4", "A1,A2")),  #
Include both behavior types
  age = mean(wcgs$age)
)


# Add predicted probabilities
newdata_int$pred_prob <- predict(model_interaction, newdata = newdata_int, type
= "response")
```

```r
# Plot the interaction
ggplot(newdata_int, aes(x = smoke,
                        y = pred_prob,
                        group = factor(dibpat, labels = c("Type B", "Type A")),
                        color = factor(dibpat, labels = c("Type B", "Type
A")))) +
  geom_line() +
  geom_point(size = 3) +
  labs(title = "Predicted Probability of CHD by Smoking and Behavior Pattern",
       x = "Smoking Status",
       y = "Predicted Probability of CHD") +
  theme_minimal()
```

Question 4

QUESTION #4

This question is based on the Western Collaborative Group Study (WCGS) dataset final-wcgs.dta considered in the previous question, and focuses on a subsidiary analysis of the relationship between smoking and CHD risk based on the quantitative variable ncigs (number of cigarettes smoked per day) and the categorical indicator npacks of number of packs smoked per day.

Question 4.1: Fit a logistic regression model for the relationship between the CHD outcome chd69 and ncigs, including age (as a linear term) and behavior pattern (dibpat) as adjustment variables.

i. Summarize evidence for a linear association between the log odds of CHD and ncigs.
ii. Discuss any limitations of the linear model fitted in part (i). (Consider the relative frequency of observed values of ncigs in your answer.)
iii. Conduct additional analyses to evaluate the plausibility of a nonlinear association between ncigs and the log odds of CHD, and write a brief summary of your findings.

Question 4.2: Fit a logistic regression model for the relationship between the CHD outcome chd69 and the categorical variable npacks, again including age and dibpat as adjustment variables.

i. Summarize the overall effect of smoking (as measured by npacks) on the odds of CHD using a likelihood ratio test.
ii. Perform a test for linear trend of the association between and the log odds of CHD and npacks, and summarize the results.
iii. Discuss briefly the limitations/ merits of the analysis conducted in question 4.2 compared to the analysis conducted in question 4.1. You may also suggest alternative analyses.

```
wcgs <- read.dta13("/Users/anyadecarlo/biostats_208/data/final-wcgs.dta",
nonint.factors = TRUE)

# Fit logistic regression model

fit_linear <- glm(chd69 ~ ncigs + age + dibpat, data = wcgs, family =
"binomial")

jtools::summ(fit_linear, confint = TRUE, digits = 3, exp = TRUE)
```

Observations	3154
Dependent variable	chd69
Type	Generalized linear model
Family	binomial

Link	logit

χ^2(3)	103.964
p	0.000
Pseudo-R^2 (Cragg-Uhler)	0.075
Pseudo-R^2 (McFadden)	0.058
AIC	1685.279
BIC	1709.505

	exp(Est.)	2.5%	97.5 %	z val.	p
(Intercept)	0.001	0.000	0.004	-11.501	0.000
ncigs	1.022	1.014	1.030	5.295	0.000
age	1.074	1.050	1.098	6.192	0.000
dibpatA1,A2	2.094	1.584	2.767	5.191	0.000

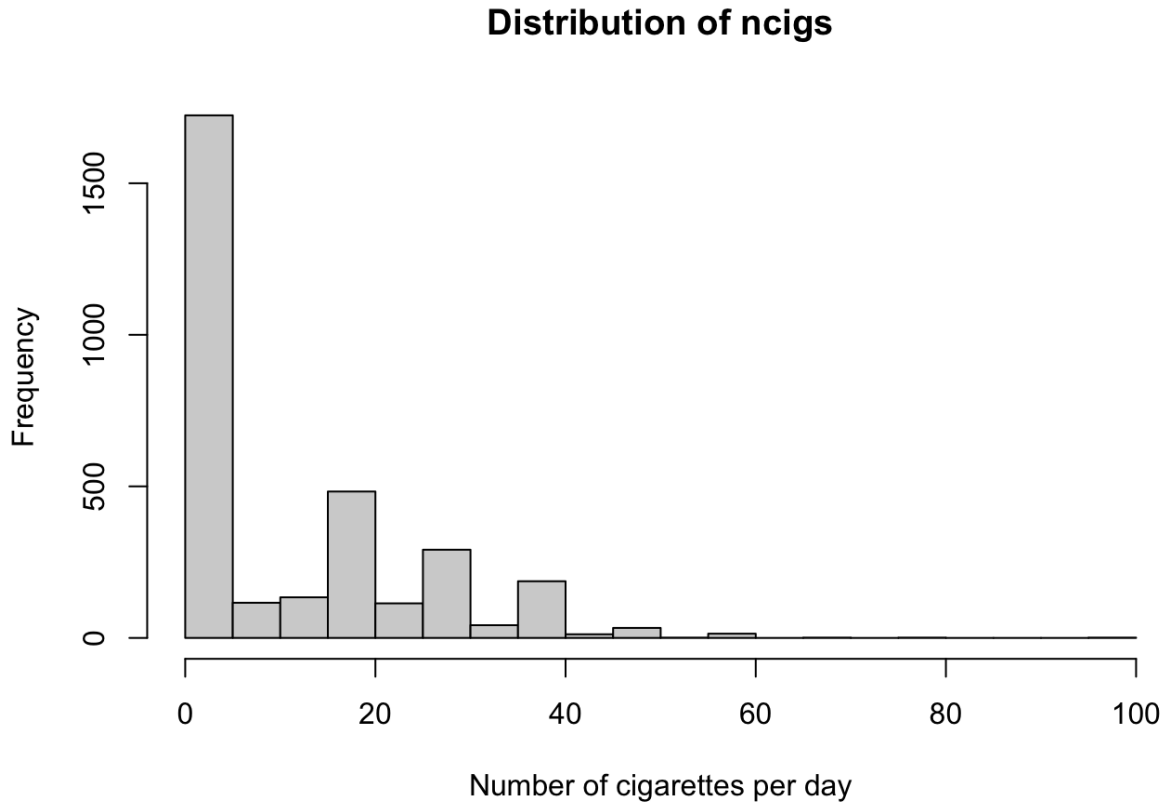
Standard errors: MLE

```
# Examine distribution of ncigs
table(wcgs$ncigs)
```

```
##
```

```
##    0    1    2    3    4    5    6    7    8    9   10   11   12   13   14
15
## 1652    7   15   14   13   23   14   12    4    3   83    3   22    4    3
102
##   16   17   18   20   22   23   24   25   27   28   30   31   33   35   36
37
##    4   13   13  453    3    5    1  105    6    2  283    2    4   36    2
2
##   38   40   42   45   48   50   55   60   70   80   99
##    1  182    1   11    1   32    1   14    1    1    1
```

```
hist(wcgs$ncigs, breaks = 20, main = "Distribution of ncigs", xlab = "Number of
cigarettes per day")
```




```
# Calculate relative frequency of ncigs
relative_freq <- table(wcgs$ncigs) / length(wcgs$ncigs)
print(relative_freq)
```

```
##
##           0           1           2           3           4
5
```

```
## 0.5237793278 0.0022194039 0.0047558656 0.0044388079 0.0041217502
0.0072923272

##            6            7            8            9           10
11

## 0.0044388079 0.0038046925 0.0012682308 0.0009511731 0.0263157895
0.0009511731

##           12           13           14           15           16
17

## 0.0069752695 0.0012682308 0.0009511731 0.0323398859 0.0012682308
0.0041217502

##           18           20           22           23           24
25

## 0.0041217502 0.1436271401 0.0009511731 0.0015852885 0.0003170577
0.0332910590

##           27           28           30           31           33
35

## 0.0019023462 0.0006341154 0.0897273304 0.0006341154 0.0012682308
0.0114140774

##           36           37           38           40           42
45

## 0.0006341154 0.0006341154 0.0003170577 0.0577045022 0.0003170577
0.0034876347

##           48           50           55           60           70
80

## 0.0003170577 0.0101458465 0.0003170577 0.0044388079 0.0003170577
0.0003170577

##           99

## 0.0003170577
```

```
# Test for linear association between ncigs and CHD

# Extract the coefficient for ncigs

ncigs_coef <- coef(summary(fit_linear))["ncigs", ]

cat("Linear association between log odds of CHD and ncigs (per cigarette):",

    round(ncigs_coef[1], 3),

    ", p-value:",

    round(ncigs_coef[4], 3))
```

```
## Linear association between log odds of CHD and ncigs (per cigarette): 0.022
, p-value: 0
```

```
# Get odds ratio and 95% CI

or_ncigs <- exp(ncigs_coef[1])

ci_lower <- exp(ncigs_coef[1] - 1.96 * ncigs_coef[2])
```

```
ci_upper <- exp(ncigs_coef[1] + 1.96 * ncigs_coef[2])
cat("\nOdds ratio per cigarette:", round(or_ncigs, 3),
    "95% CI: [", round(ci_lower, 3), "-", round(ci_upper, 3), "]")
##
## Odds ratio per cigarette: 1.022 95% CI: [ 1.014 - 1.03 ]
# Create restricted cubic spline for ncigs
# Define knots with meaningful smoking quantities
knots <- c(10, 20, 30, 40)
wcgs$rcspline <- rcspline.eval(wcgs$ncigs, knots = knots)


# Fit model with splines
fit_nonlinear <- glm(chd69 ~ rcspline + age + dibpat, data = wcgs, family =
"binomial")
jtools::summ(fit_nonlinear, confint = TRUE, digits = 3)
```

Observations	3154
Dependent variable	chd69
Type	Generalized linear model
Family	binomial
Link	logit

$\chi^2(4)$	102.358
p	0.000
Pseudo-R^2 (Cragg-Uhler)	0.074
Pseudo-R^2 (McFadden)	0.057

AIC	1688.886
BIC	1719.168

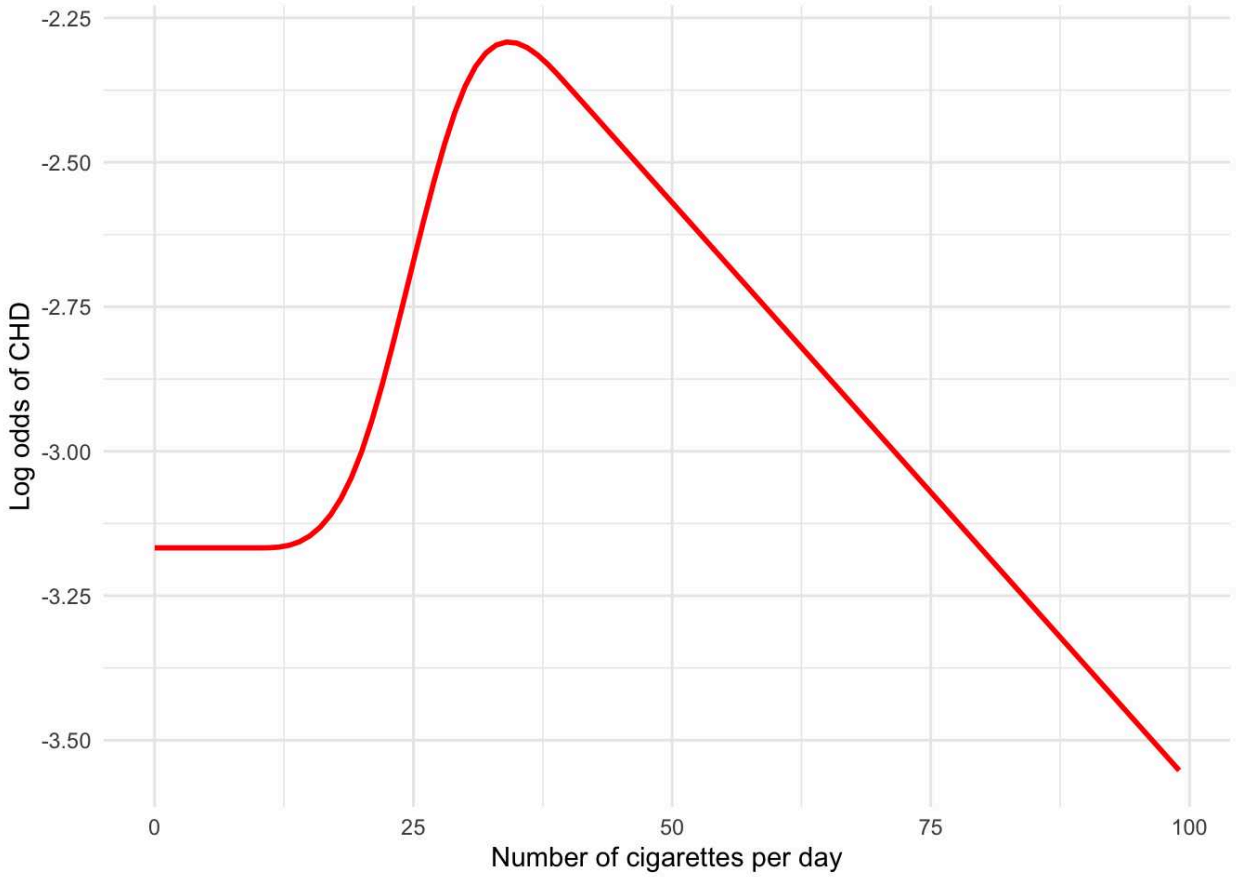
	Est.	2.5%	97.5%	z val.	p
(Intercept)	-6.502	-7.613	-5.391	-11.469	0.000
rcspline1	0.150	0.084	0.216	4.467	0.000
rcspline2	-0.480	-0.718	-0.241	-3.945	0.000
age	0.072	0.049	0.095	6.258	0.000
dibpatA1,A2	0.751	0.472	1.030	5.275	0.000

Standard errors: MLE

```
# Test for nonlinearity (joint test of all spline terms except the first)
anova(fit_linear, fit_nonlinear, test = "Chisq")
## Analysis of Deviance Table
##
## Model 1: chd69 ~ ncigs + age + dibpat
## Model 2: chd69 ~ rcspline + age + dibpat
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      3150     1677.3
## 2      3149     1678.9  1  -1.6065
# Visualize the relationship
```

```
new_data <- data.frame(
  ncigs = seq(0, max(wcgs$ncigs), length.out = 100),
  age = mean(wcgs$age),
  dibpat = "B3,B4"  # Using Type B behavior as reference
)
new_data$rcspline <- rcspline.eval(new_data$ncigs, knots = knots)
new_data$predicted_log_odds <- predict(fit_nonlinear, newdata = new_data)

# Plot the relationship
ggplot(new_data, aes(x = ncigs, y = predicted_log_odds)) +
  geom_line(color = "red", linewidth = 1) +
  labs(x = "Number of cigarettes per day", y = "Log odds of CHD") +
  theme_minimal()
```




```
# Question 4.2 (i): Fit full model with npacks
fit_full <- glm(chd69 ~ npacks + age + dibpat, data = wcgs, family =
"binomial")
```

```
jtools::summ(fit_full, confint = TRUE, digits = 3, exp = TRUE)
```

Observations	3154
Dependent variable	chd69
Type	Generalized linear model
Family	binomial
Link	logit

χ^2(6)	105.266
p	0.000
Pseudo-R^2 (Cragg-Uhler)	0.076
Pseudo-R^2 (McFadden)	0.059
AIC	1689.978
BIC	1732.373

	exp(Est.)	2.5%	97.5 %	z val.	p
(Intercept)	0.001	0.000	0.004	-11.505	0.000
npacks	1.073	0.481	2.394	0.172	0.864

	exp(Est.)	2.5%	97.5 %	z val.	p
npacks1/2-1	1.542	1.104	2.153	2.541	0.011
npacks>1-2	2.151	1.563	2.959	4.705	0.000
npacks>2	3.158	1.606	6.212	3.332	0.001
age	1.074	1.050	1.098	6.183	0.000
dibpatA1,A2	2.103	1.591	2.780	5.218	0.000

Standard errors: MLE

```
# Fit reduced model without npacks
fit_reduced <- glm(chd69 ~ age + dibpat, data = wcgs, family = "binomial")
jtools::summ(fit_reduced, confint = TRUE, digits = 3, exp = TRUE)
```

Observations	3154
Dependent variable	chd69
Type	Generalized linear model
Family	binomial
Link	logit

$\chi^2(2)$	77.083

p	0.000
Pseudo-R^2 (Cragg-Uhler)	0.056
Pseudo-R^2 (McFadden)	0.043
AIC	1710.160
BIC	1728.330

	exp(Est.)	2.5%	97.5 %	z val.	p
(Intercept)	0.002	0.001	0.006	-11.086	0.000
age	1.071	1.047	1.095	6.040	0.000
dibpatA1,A2	2.224	1.686	2.934	5.658	0.000

Standard errors: MLE

```
# Likelihood ratio test
anova(fit_reduced, fit_full, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: chd69 ~ age + dibpat
## Model 2: chd69 ~ npacks + age + dibpat
##   Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
## 1      3151     1704.2
## 2      3147     1676.0  4   28.182 1.146e-05 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Question 4.2 (ii): Test for linear trend

# Approach 1: Using test for trend for binary outcome
# First create a contingency table of CHD by npacks
tab <- table(wcgs$chd69, wcgs$npacks)
# Calculate column percentages
col_percentages <- prop.table(tab, margin = 2) * 100
print(addmargins(col_percentages)) # proportions
```

```
##
##                0       <1/2      1/2-1       >1-2         >2        Sum
##   no    94.067797  93.333333  91.000000  87.854890  80.952381 447.208400
##   yes    5.932203   6.666667   9.000000  12.145110  19.047619  52.791600
##   Sum 100.000000 100.000000 100.000000 100.000000 100.000000 500.000000
```

```
# Test for trend using prop.trend.test
# Extract the counts of CHD for each npacks level
chd_counts <- tab["yes", ]
# Get the total counts for each npacks level
total_counts <- colSums(tab)
# Run the trend test
prop.trend.test(chd_counts, total_counts)
```

```
##
##  Chi-squared Test for Trend in Proportions
##
## data:  chd_counts out of total_counts ,
##  using scores: 1 2 3 4 5
## X-squared = 32.226, df = 1, p-value = 1.373e-08
```

```
# Approach 2: Using logistic regression to test for linear trend
# Create a model with npacks as a continuous variable
# First check if npacks is a factor
class(wcgs$npacks)
```

```
## [1] "factor"
```

```
# Create a numeric score for npacks if needed
wcgs$npacks_score <- as.numeric(wcgs$npacks)
```

```r
# Test for trend by treating npacks as a continuous predictor
fit_trend <- glm(chd69 ~ npacks_score + age + dibpat, data = wcgs, family = "binomial")
jtools::summ(fit_trend, confint = TRUE, digits = 3)
```

Observations	3154
Dependent variable	chd69
Type	Generalized linear model
Family	binomial
Link	logit

χ^2(3)	104.639
p	0.000
Pseudo-R^2 (Cragg-Uhler)	0.076
Pseudo-R^2 (McFadden)	0.059
AIC	1684.604
BIC	1708.830

	Est.	2.5%	97.5%	z val.	p
(Intercept)	-6.833	-7.970	-5.697	-11.782	0.000

	Est.	2.5%	97.5 %	z val.	p
npacks_score	0.259	0.162	0.355	5.249	0.000
age	0.071	0.048	0.094	6.171	0.000
dibpatA1,A2	0.748	0.470	1.027	5.264	0.000

Standard errors: MLE

```
# Test for departure from linearity by comparing models
# This compares the model with npacks as continuous vs categorical
fit_cat <- glm(chd69 ~ factor(npacks) + age + dibpat, data = wcgs, family =
"binomial")
anova(fit_trend, fit_cat, test = "Chisq")
## Analysis of Deviance Table
##
## Model 1: chd69 ~ npacks_score + age + dibpat
## Model 2: chd69 ~ factor(npacks) + age + dibpat
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1      3150     1676.6
## 2      3147     1676.0  3  0.62611   0.8904
# Test for linearity
# Set polynomial contrasts for npacks
wcgs$npacks_ord <- wcgs$npacks  # make a copy to preserve factor levels
contrasts(wcgs$npacks_ord) <- contr.poly(nlevels(wcgs$npacks_ord))


# Fit logistic regression with contrasts
fit_contrast <- glm(chd69 ~ npacks_ord + age + dibpat, data = wcgs, family =
"binomial")
jtools::summ(fit_contrast, confint = TRUE, digits = 3)
```

Observations	3154
Dependent variable	chd69
Type	Generalized linear model
Family	binomial
Link	logit

$\chi^2(6)$	105.266
p	0.000
Pseudo-R^2 (Cragg-Uhler)	0.076
Pseudo-R^2 (McFadden)	0.059
AIC	1689.978
BIC	1732.373

	Est.	2.5%	97.5%	z val.	p
(Intercept)	-6.074	-7.191	-4.957	-10.661	0.000
npacks_ord.L	0.947	0.448	1.447	3.718	0.000

	Est.	2.5%	97.5 %	z val.	p
npacks_ord.Q	0.160	-0.284	0.604	0.706	0.480
npacks_ord.C	-0.076	-0.633	0.480	-0.268	0.788
npacks_ord^4	0.048	-0.391	0.487	0.215	0.830
age	0.071	0.049	0.094	6.183	0.000
dibpatA1,A2	0.743	0.464	1.023	5.218	0.000

Standard errors: MLE

```
# Question 4.2 (iii): Discussion of limitations/merits compared to analysis in 4.1


# Calculate the percent increase in odds for each additional pack

# Using the trend model

beta_trend <- coef(fit_trend)["npacks_score"]

percent_increase <- (exp(beta_trend) - 1) * 100

cat("For each additional pack of cigarettes per day, the odds of CHD increase by approximately",

    round(percent_increase, 1), "% (95% CI:",

    round((exp(confint(fit_trend)["npacks_score",1]) - 1) * 100, 1), "% to",

    round((exp(confint(fit_trend)["npacks_score",2]) - 1) * 100, 1), "%)\n\n")
```

```
## Waiting for profiling to be done...

## Waiting for profiling to be done...

## For each additional pack of cigarettes per day, the odds of CHD increase by approximately 29.5 % (95% CI: 17.6 % to 42.7 %)
```

```
# Compare AIC values for the continuous ncigs model vs categorical npacks model
```

```r
cat("AIC for continuous ncigs model:", AIC(fit_linear), "\n")
```

```
## AIC for continuous ncigs model: 1685.279
```

```r
cat("AIC for categorical npacks model:", AIC(fit_full), "\n\n")
```

```
## AIC for categorical npacks model: 1689.978
```

```r
# Compare distribution of npacks vs ncigs
table(wcgs$npacks)
```

```
##
##     0  <1/2 1/2-1  >1-2    >2
##  1652   105   700   634    63
```

```r
summary(wcgs$ncigs)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     0.0     0.0     0.0    11.6    20.0    99.0
```

```r
# Question 4.3: Interpretation of results

# Create a summary table of the odds ratios for npacks categories
or_table <- exp(coef(fit_full))
or_ci <- exp(confint(fit_full))
```

```
## Waiting for profiling to be done...
```

```r
# Combine into a data frame for easier viewing
or_results <- data.frame(
  Variable = names(or_table),
  OR = or_table,
  Lower_CI = or_ci[,1],
  Upper_CI = or_ci[,2]
)

# Display only the npacks categories
npacks_results <- or_results[grep("npacks", or_results$Variable),]
print(npacks_results)
```

```
##                Variable       OR  Lower_CI Upper_CI
## npacks<1/2   npacks<1/2 1.072812 0.4391741 2.239714
## npacks1/2-1 npacks1/2-1 1.541657 1.0998105 2.147059
## npacks>1-2   npacks>1-2 2.150843 1.5603123 2.956258
## npacks>2       npacks>2 3.158293 1.5416040 6.029831
```

```r
# Calculate the percent increase in odds for each additional pack

# Using the trend model

beta_trend <- coef(fit_trend)["npacks_score"]

percent_increase <- (exp(beta_trend) - 1) * 100

cat("For each additional pack of cigarettes per day, the odds of CHD increase
by approximately",

    round(percent_increase, 1), "% (95% CI:",

    round((exp(confint(fit_trend)["npacks_score",1]) - 1) * 100, 1), "% to",

    round((exp(confint(fit_trend)["npacks_score",2]) - 1) * 100, 1), "%)")
```

```
## Waiting for profiling to be done...

## Waiting for profiling to be done...

## For each additional pack of cigarettes per day, the odds of CHD increase by
approximately 29.5 % (95% CI: 17.6 % to 42.7 %)
```

Index of comments

2.1 -.25 do not interpret the estimate in the test of linear trend. inference should only focus on the hypothesis since the estimate is just the associated test statistic, not a slope coefficient.

3.1 -.5 Describe an overall F test for physact which focuses on the overall effect, rather than the single significant effect estimate. Also incorrect interpretation of test for linear trend (see previous comment)

6.1 5/5

7.1 5/5

8.1 4/6

9.1 6/6

13.1 4/4

14.1 5/5

15.1 4/4

16.1 4/4

17.1 4/4

17.2 4/4

18.1 5/5